Effects of N-3 on metabolic syndrome: Supplementary data

...

Contents

1 Introduction 5

1.1 Project organisation **5**

1.1.1 Chapters 5

1.1.2 R scripts 5

1.1.3 Stan models 5

1.1.4 Reproducibility 5

2 Analytic approach and data description 7

2.1 Data analysis **7**

2.2 Description of the data set and participants **7**

3 Effects of N-3 supplementation of serum fatty acids 11

3.0.1 Model expansion: Heterogeneous variance 24

3.1 Inference **28**

4 Effects of N-3 supplementation on indications of metabolic syndrome - Introduction 29

4.1 Effects of N-3 supplementation on indicies of metabolic syndrome **29**

4.1.1 Synthetic data 37

Bibliography 45

1. Introduction

This document includes supplementary data and computational details related to the paper "Effects of ...".

1.1 Project organisation

This notebook is built from source code available at the github repository trainome/alpha-omega-trial.

1.1.1 Chapters

Each chapter contains model definitions and checks for analyses in the project.

1.1.2 R scripts

- `R/fig-opt-fun.R` - Helper functions and settings for data visualisations.
- `R/helpers.R` - Helper functions for modelling.
- `R/packages.R` - Loading required packages.

1.1.3 Stan models

All Stan model scripts are stored in `/models/`. If model code is executed locally, model executables are also stored here. Model fits are stored in `fits/` but not pushed to github due to their large size.

1.1.4 Reproducibility

The project uses renv for a snapshot

2. Analytic approach and data description

2.1 Data analysis

Descriptive data are presented as mean and standard deviation (SD). For inferential statistics, Bayesian models were constructed using the Stan probabilistic programming language (1) and fitted using the `cmdstanr` interface (2) in R (3). All sampling from prior and posterior distributions were done using the Hamiltonian Monte Carlo (HMC) no-U-turn sampler (Stan Reference Manual: MCMC sampling). Performance of the HMC algorithm was assessed through chain convergence statistics (R-hat) and visually inspecting convergence and stationary sampling across multiple HMC chains. A general workflow was adopted where models were incrementally built and tested for validity through simulations. First, implications of prior distributions were evaluated using prior-predictive simulations. Priors were selected to constrain model predictions and parameter values to *a priori* plausible ranges. This approach encourages the use of weakly informative priors on model parameters, which regularize models and shrink estimates and predictions without distorting the data's influence on inference (4–6). In relevant cases, synthetic data were used to confirm that model inferences were not biased by model misspecification. Samples from the joint posterior distribution were used to create posterior predictive checks, in which model implications are simulated and compared with characteristics of the observed data (6). Selected models were also assessed for inferential capabilities using synthetic data simulations where parameter values are known. The sensitivity of model specifications to inference was assessed by comparing predictions across different models.

2.2 Description of the data set and participants

The overall objective of the present analysis is to determine the effect of N-3 supplementation in combination with resistance training on glucose metabolism and indicators of metabolic health. Outcome variables in the analysis are:

- Blood glucose in response to a oral glucose tolerance-test
- Metabolic syndrome (derived from sub-scores)
  - Triglycerides
  - Waist circumference
  - Blood pressure
  - HDL Cholesterol
  - Resting glucose
- Body mass index
- Relative fat mass
- HOMA2-IR
- Adiponectin
- Red blood cell fatty acid composition

A total of 147 participants are included in the data analysis. Participants were randomized into two groups, an active treatment-group (N-3 fatty acids, XX mg day^{-1}) and a placebo control group (Placebo; Table 2.1).

Table 2.1: Participant characteristics.

	N-3(♂ n = 32; ♀ n = 42)	Placebo(♂ n = 35; ♀ n = 38)
	Male	
Age (years)	47.3 (7.7) [29.0 - 60.0]	48.5 (5.8) [36.0 - 59.0]
Stature (cm)	180.5 (8.1) [153.3 - 193.2]	180.9 (6.4) [165.2 - 191.7]
Body mass (kg)	95.5 (17.1) [68.6 - 151.9]	97.6 (17.6) [71.2 - 142.2]
Body mass index ($kg \times m^{-2}$)	29.4 (5.1) [20.5 - 45.1]	30.0 (6.1) [23.1 - 52.1]
	Female	
Age (years)	44.5 (8.1) [30.0 - 59.0]	46.6 (9.4) [23.0 - 60.0]
Stature (cm)	168.6 (6.0) [158.0 - 181.9]	167.6 (6.4) [152.1 - 179.9]
Body mass (kg)	83.3 (16.6) [57.7 - 135.7]	79.3 (14.2) [52.9 - 113.8]
Body mass index ($kg \times m^{-2}$)	29.4 (5.8) [18.4 - 44.9]	28.2 (4.8) [20.4 - 39.9]

Data are presnted as Mean (SD) [Min-Max]

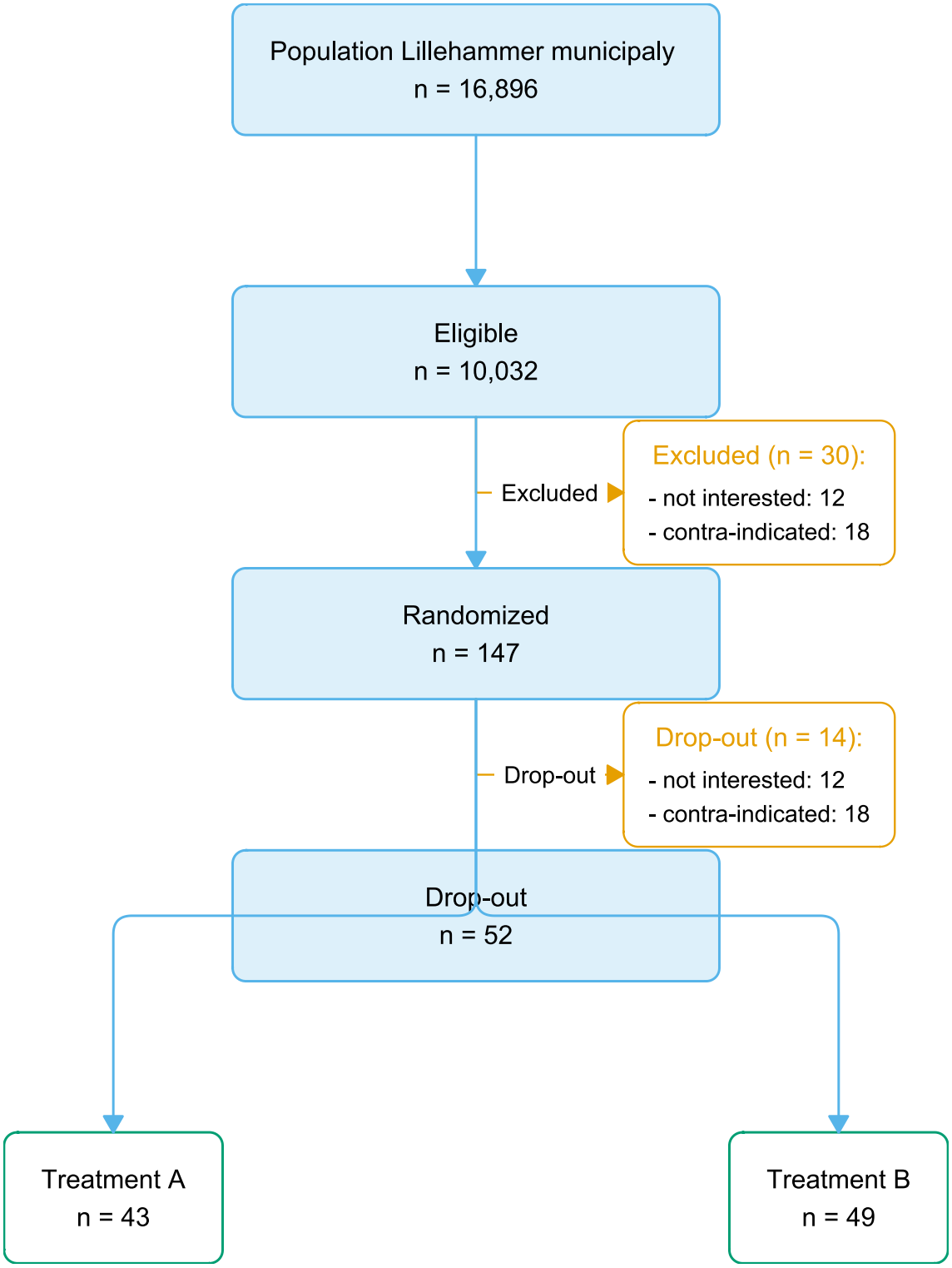


Figure 2.1: Participant recruitment, allocation and drop-out.

3. Effects of N-3 supplementation of serum fatty acids

Fatty acids, including including EPA (C205n3) and DHA (226n3) where measured in red blood cells. We will assess the effect of n-3 supplementation on group level fatty acid profiles. As preliminary models indicated heteroscedasticity of residuals, log-transformed and standardized data was used in modelling. This allows for the same weakly regularizing priors for all fatty acids (FA).

The model was designed to estimate differences between treatments (TX) at post-treatment time-points (T2 and T4) using pre-treatment ($\text{Pre}_{i[\text{FA}]}$) values as a covariate. The intercept in the model ($\alpha_{[\text{FA, TI, TX}]}$) gives us mean estimates for fatty-acid FA, time-point TI, and treatment TX.

$$
\begin{aligned}
y_{i[\text{FA}]} &\sim \text{Normal}\ (\mu_{i[\text{FA}]}, \sigma_{\text{FA}}) \\
\mu_{i[\text{FA}]} &= \alpha_{[\text{FA, TI, TX}]} + \beta_{\text{FA}} \times \text{Pre}_{i[\text{FA}]} \\
\alpha_{[\text{FA, TX}]} &\sim \text{Normal}\ (0, 1) \\
\beta_{\text{FA}} &\sim \text{Normal}\ (1, 1) \\
\sigma_{\text{FA}} &\sim \text{Exponential}\ (1)
\end{aligned}
\tag{3.1}
$$

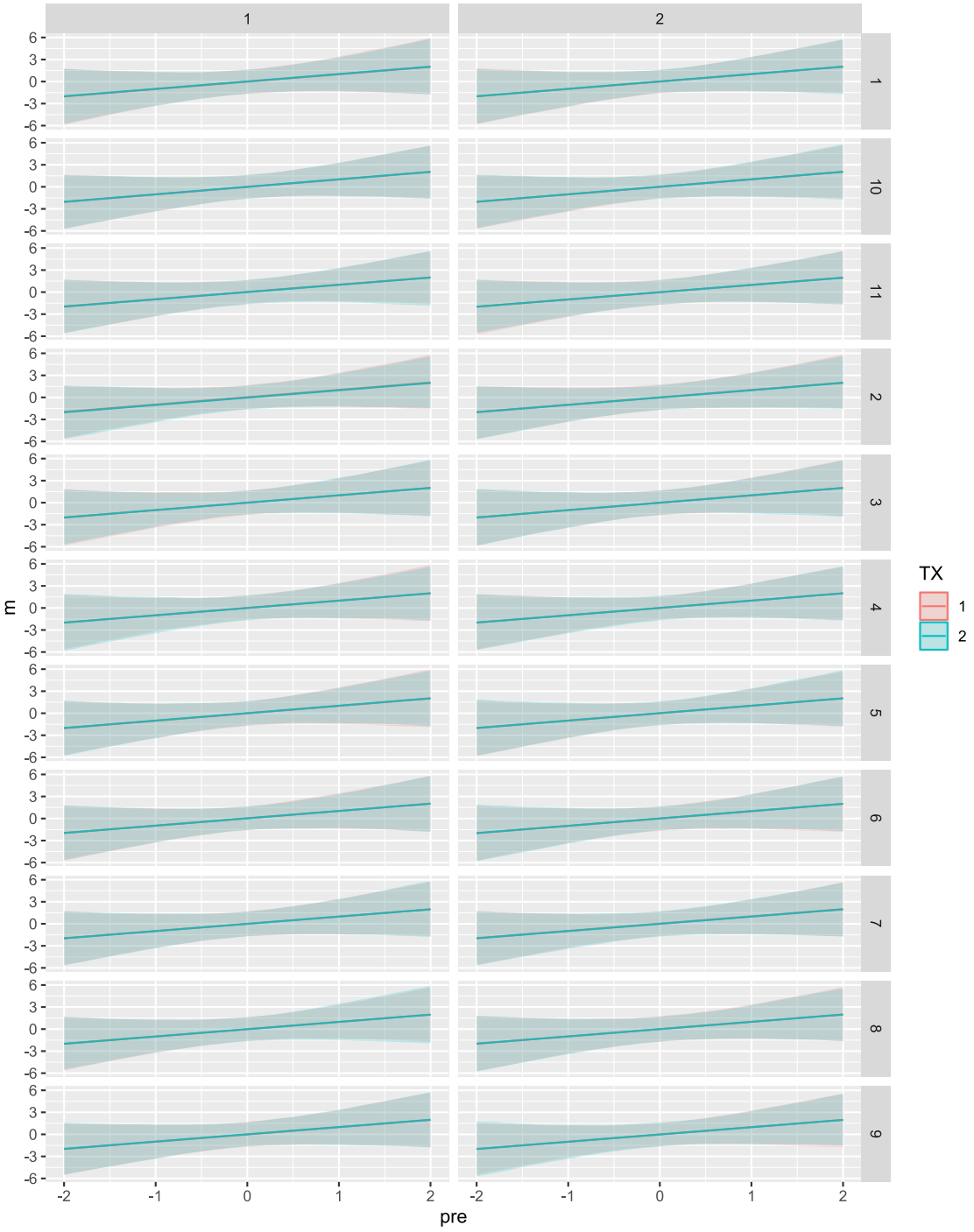


Figure 3.1: Prior predictive check for target-wise models of post-treatment fatty acid concentrations conditional on pre-treatment values. The model expects log-transformed standardized values and the priors suggest no difference between treatment groups and a proportional relationship between th pre-intervention values and post-intervention measurements.

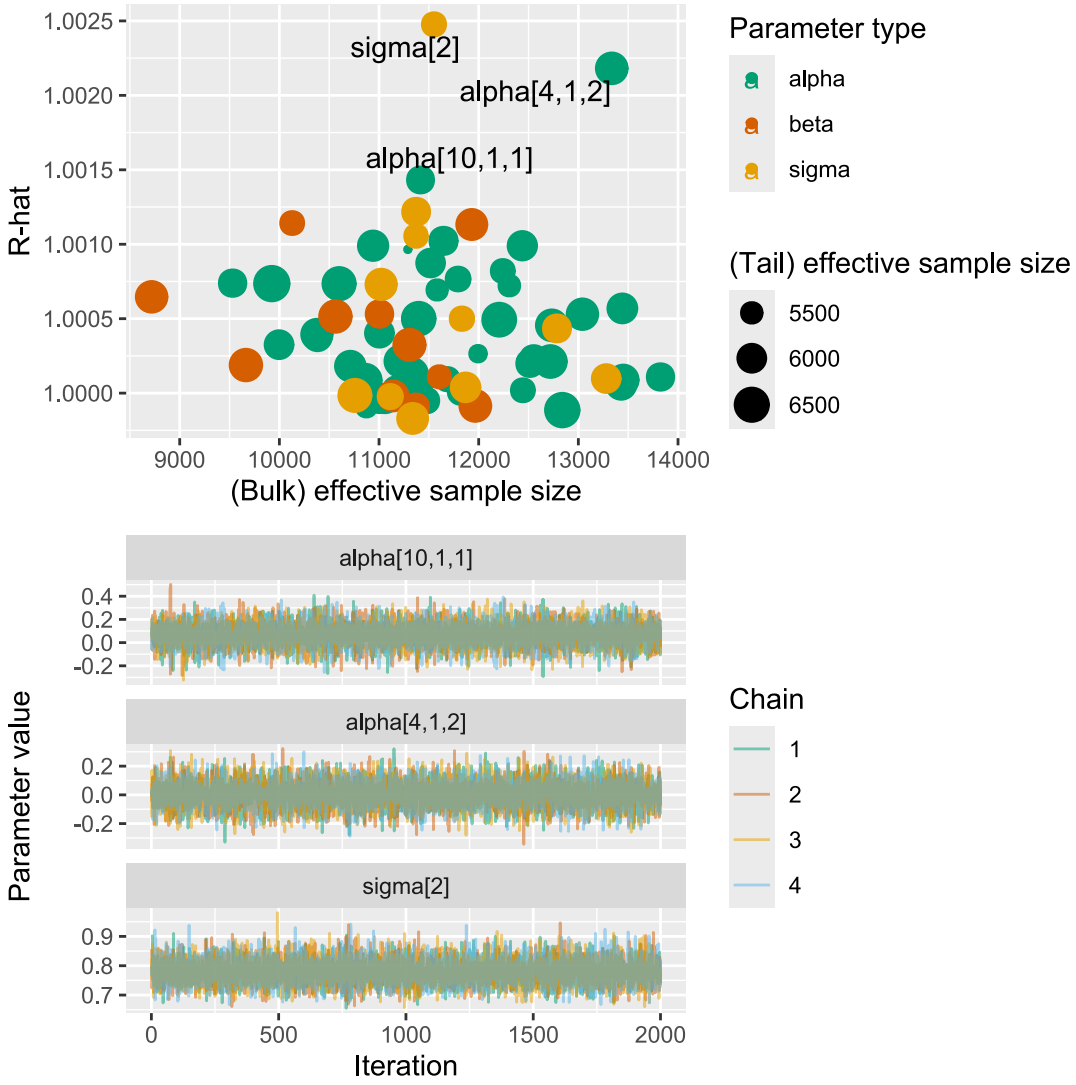


Figure 3.2: Model performance metrics from Model 2.

A first model uses only complete-cases. The prior predictive check confirms that the model expects eleven fatty acids, a positive relationship between pre-intervention values and post-treatment values, and no differences in treatments (Figure 3.1). Sampling from the posterior was non-problematic as R-hat values approached 1 and chains mixed and converged well (Figure 3.2).

Posterior predictive checks showed satisfactory patterns between replicated data (yrep; thin lines, one per draw in Figure 3.3) and observed data (thick lines in Figure 3.3).

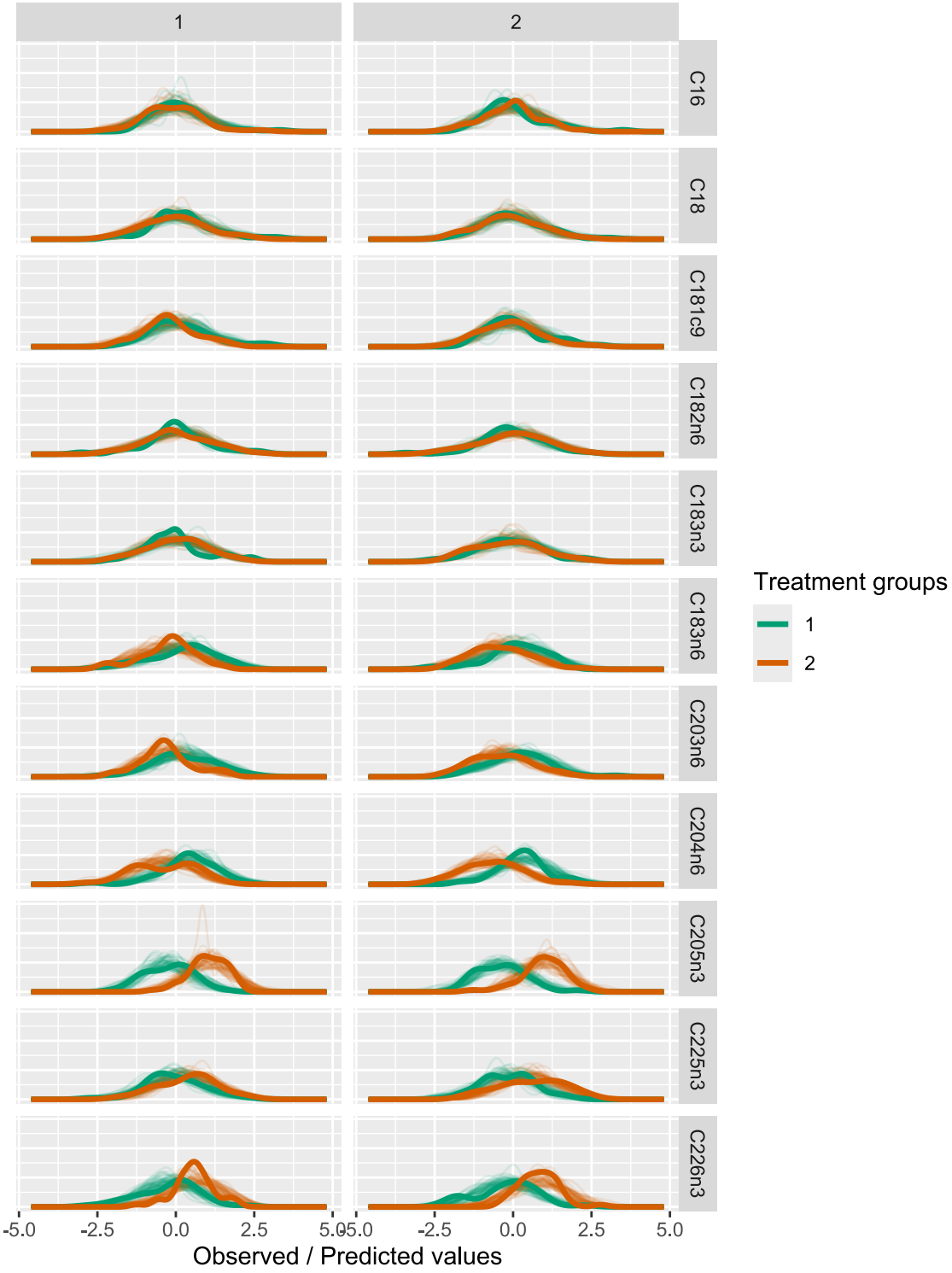


Figure 3.3: Posterior predictive check of Model 2.

Treatment with N-3 fatty acids robustly increased red blood cell fatty acid composition levels of C205n3, C225n3 and C226n3 fatty acids compared to placebo (Figure 3.4 and Figure 3.5). Post-treatment values were lower in the N-3 group for C181c9, C183n6, C203n6 and C204n6 with high probability at both T2 and T4 (Figure 3.5).

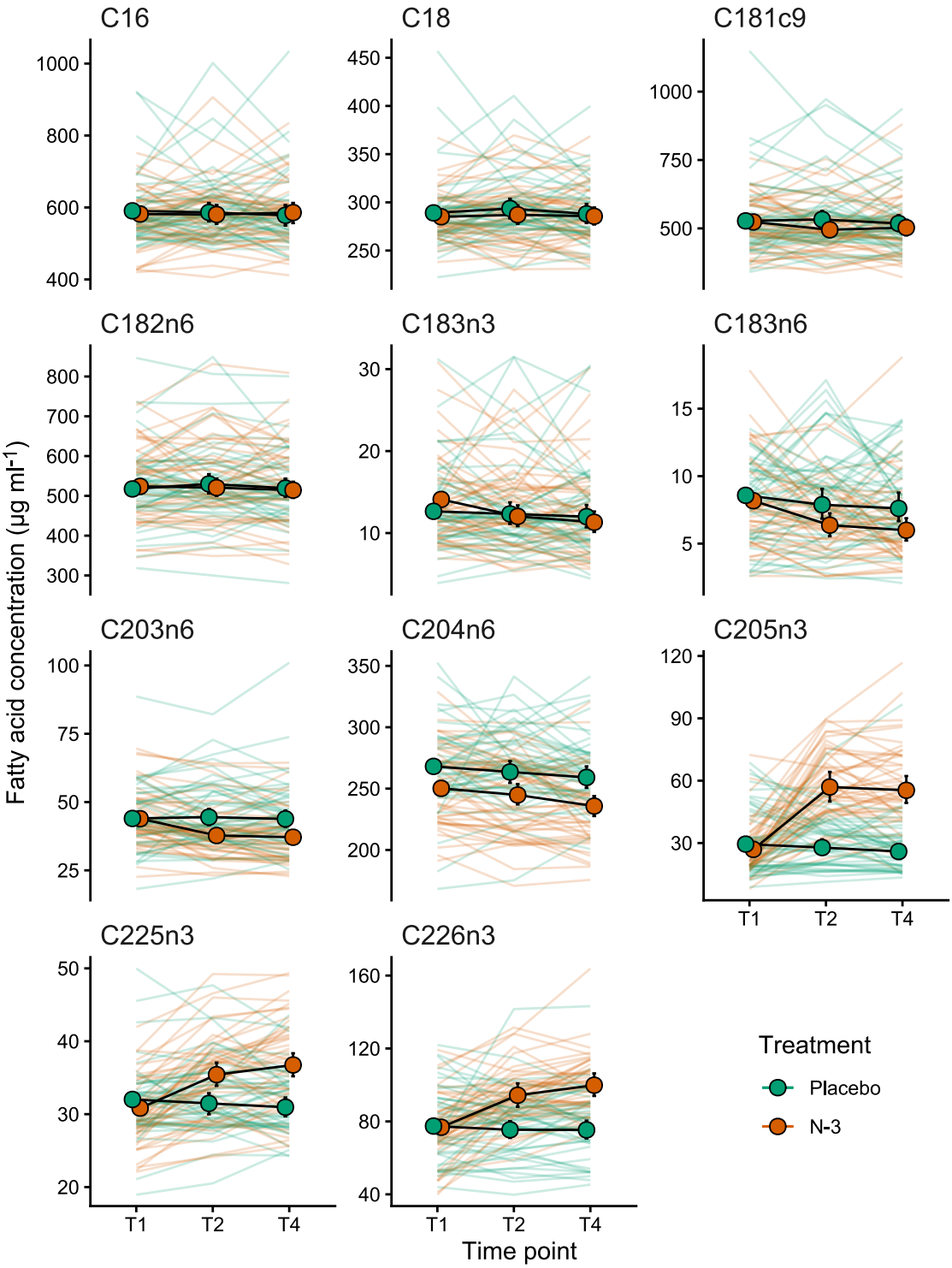


Figure 3.4: Predictions of fatty acid concentration in response to treatment. Pre-intervention values are means per treatment group, post-intervention means are predicted from the model with 99% PI.

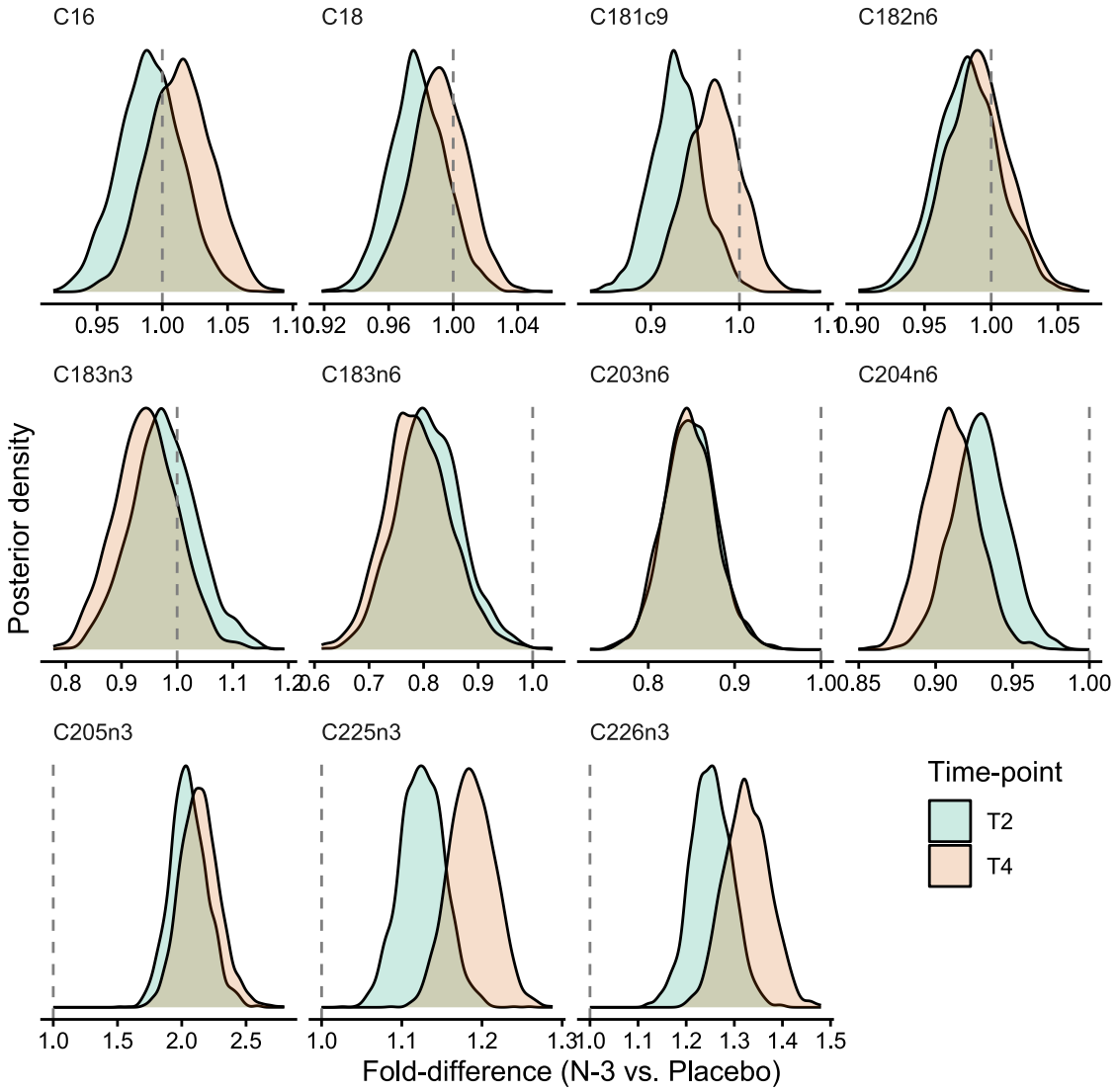


Figure 3.5: Posterior densities for differences between treatment groups in fatty acid concentrations.

Extending the first model with missing data imputations allows us to estimate the effect of treatment on fatty acid profiles with intention-to-treat (ITT) by averaging over the uncertainty associated with the missing values. Missing data imputation is done on post-treatment scores only with the total sample size being n = 143 with 14 and 36% of participants missing at time points T2 and T4, respectively.

The imputation mechanism replaces missing values with parameters estimated from observed data (pre-treatment values and treatment allocation) through the likelihood.

The imputation model is defined as:

$$
\begin{aligned}
y^{\text{all}}_{i[\text{FA}]} &\sim \text{Normal}\ (\mu_{i[\text{FA}]}, \sigma_{\text{FA}}) \\
\mu_{i[\text{FA}]} &= \alpha_{[\text{FA, TI, TX}]} + \beta_{\text{FA}} \times \text{Pre}_{i[\text{FA}]} \\
y^{\text{all}}_{i[\text{FA}]} &= \begin{cases} y^{\text{obs}}_{i[\text{FA}]} \text{ if observed} \\ y^{\text{mis}}_{i[\text{FA}]} \text{ if missing} \end{cases} \\
y^{\text{mis}}_{i[\text{FA}]} &\sim \text{Normal}\ (0, 2) \\
\alpha_{[\text{FA, TX}]} &\sim \text{Normal}\ (0, 1) \\
\beta_{\text{FA}} &\sim \text{Normal}\ (1, 1) \\
\sigma_{\text{FA}} &\sim \text{Exponential}\ (1)
\end{aligned}
\tag{3.2}
$$

Prior predictive simulations confirm the general structure of the model (Figure 3.6). Model performance metrics from posterior sampling also indicates a functioning model (Figure 3.7).

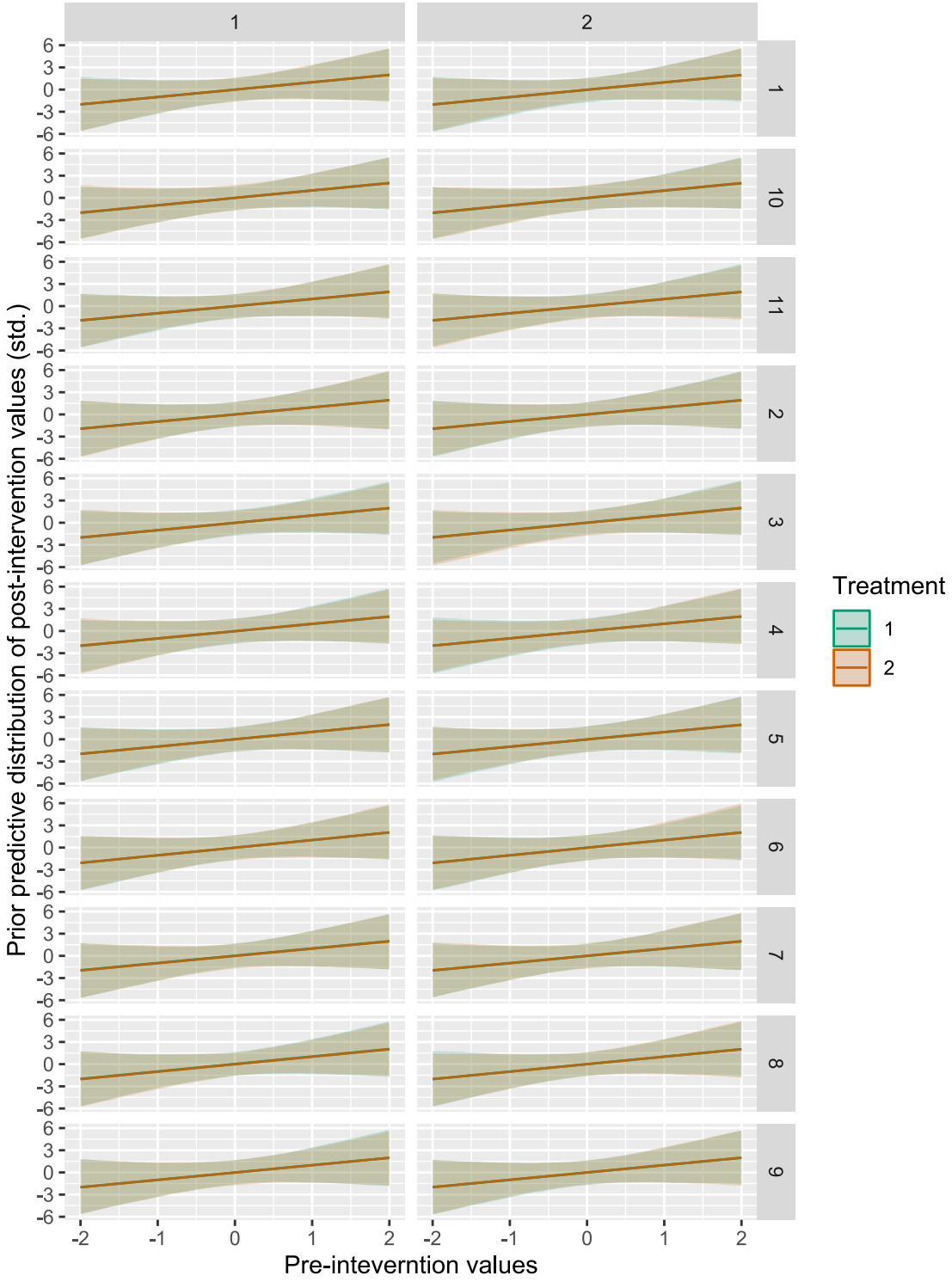


Figure 3.6: Prior predictive check model 3 (data imputation).

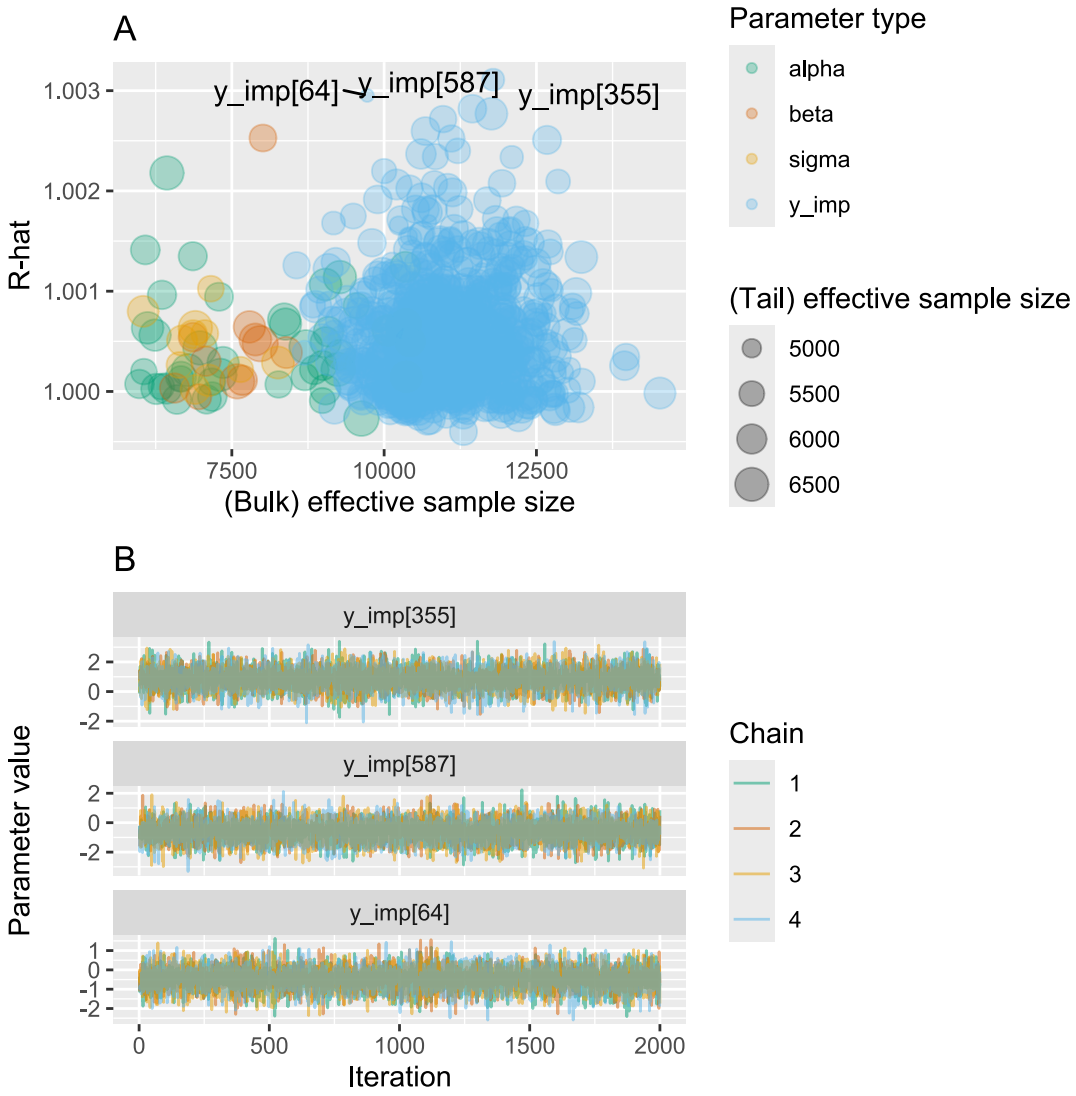


Figure 3.7: Model performance metrics model 3.

Posterior predictive checks indicated that the model capture general patterns in the data (Figure 3.8).

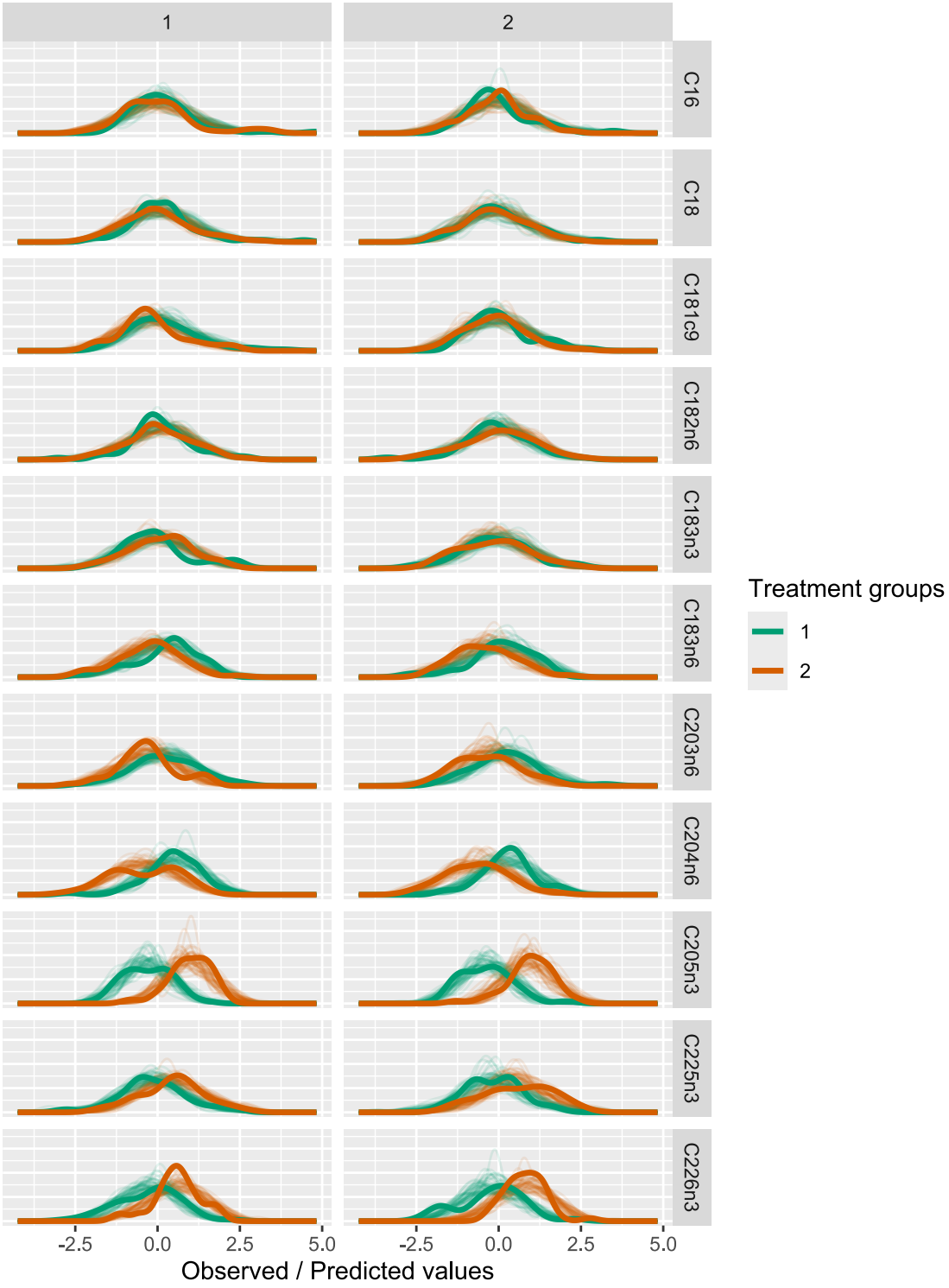


Figure 3.8: Posterior predictive check of Model 3.

The imputation model generally agreed with the complete-case analysis on the effect of treatment of serum fatty acid profiles (Figure 3.9, Figure 3.10, Figure 3.11, and Figure 3.12).

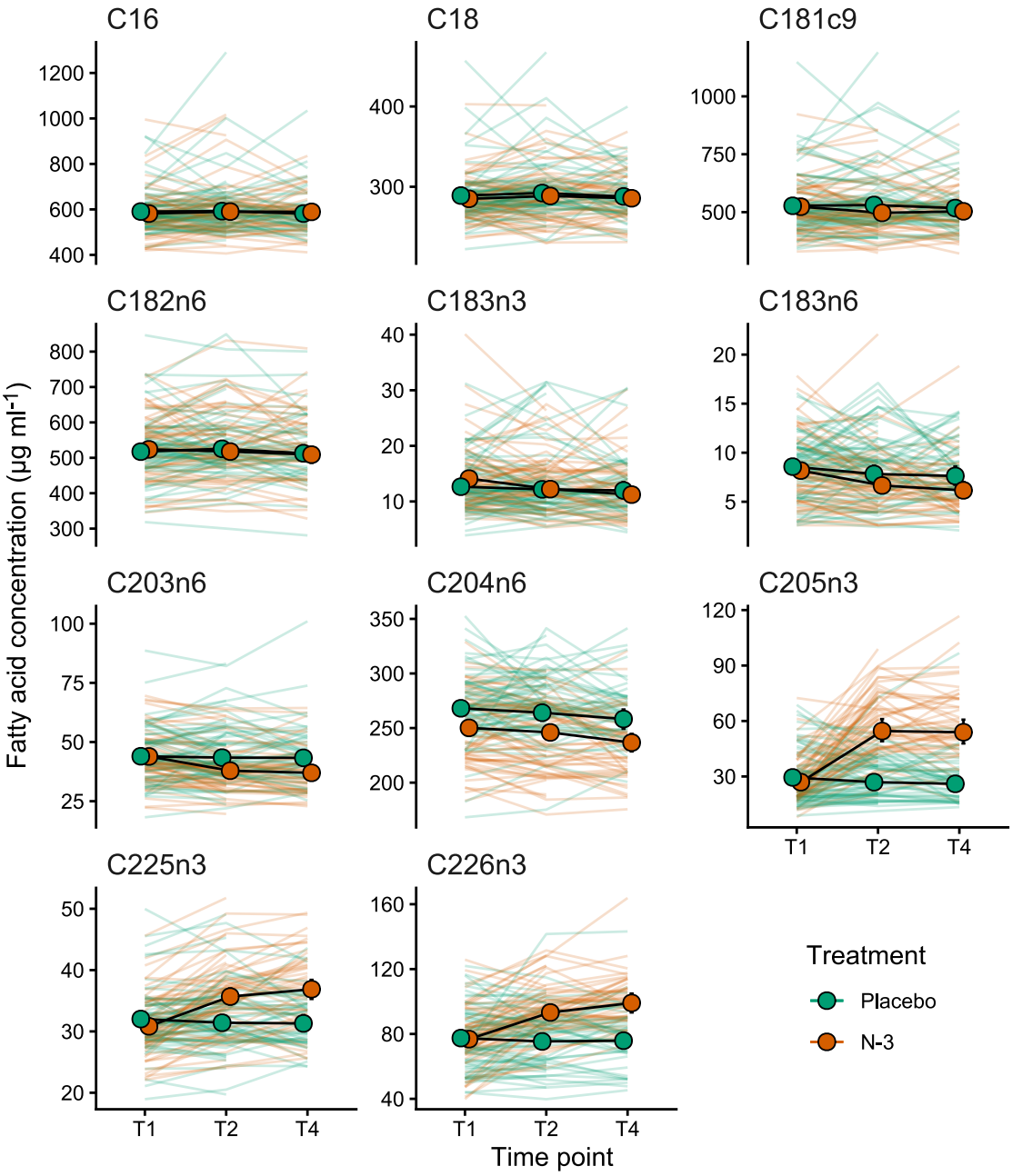


Figure 3.9: Model predictions. Pre-intervention values are observed means, post-treatment (T2, T4) values are model predictions conditional pre-intervention value with 99% percentile intervals.

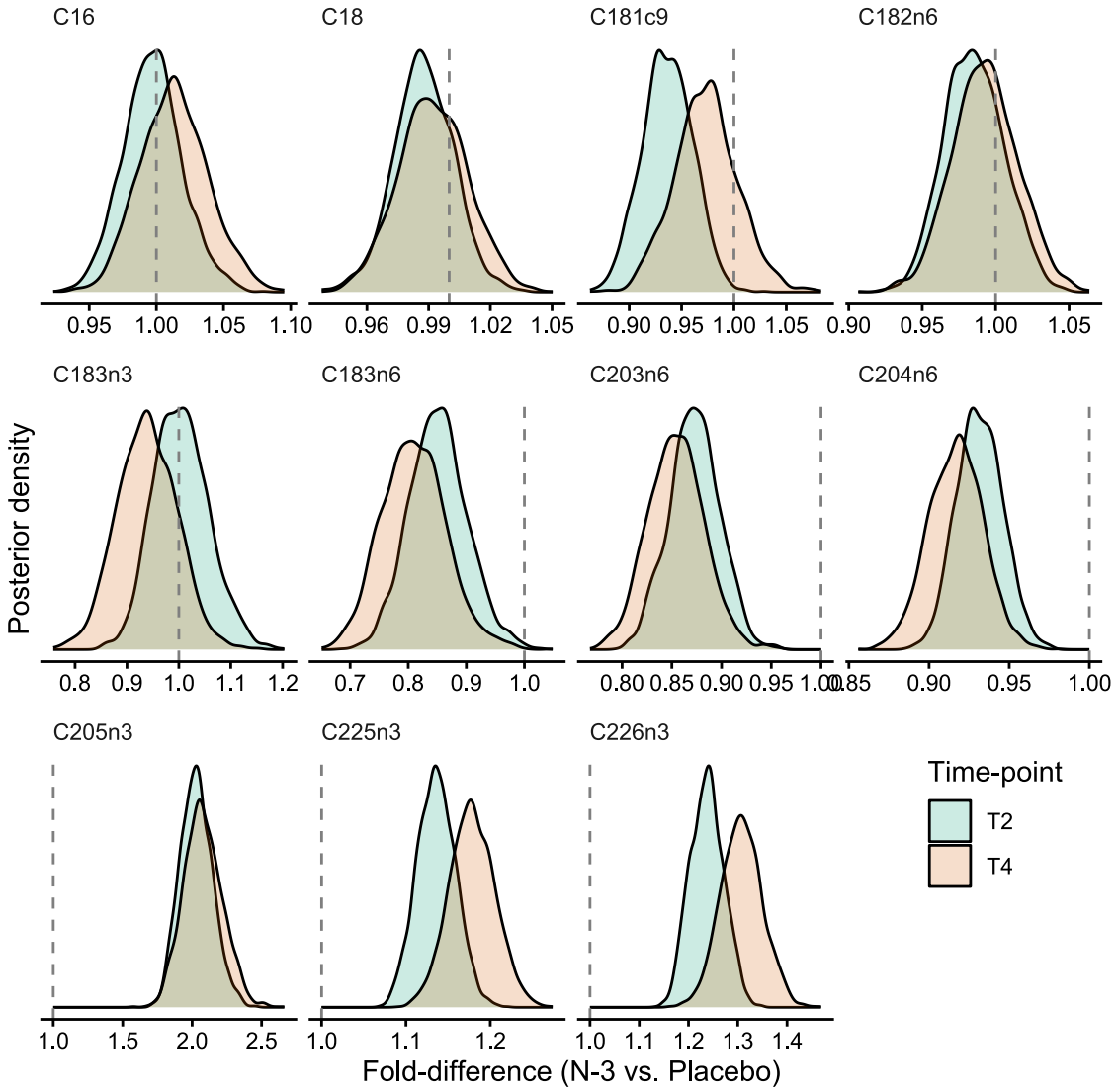


Figure 3.10: Posterior densities for differences between treatment groups in fatty acid concentrations (data imputation model).

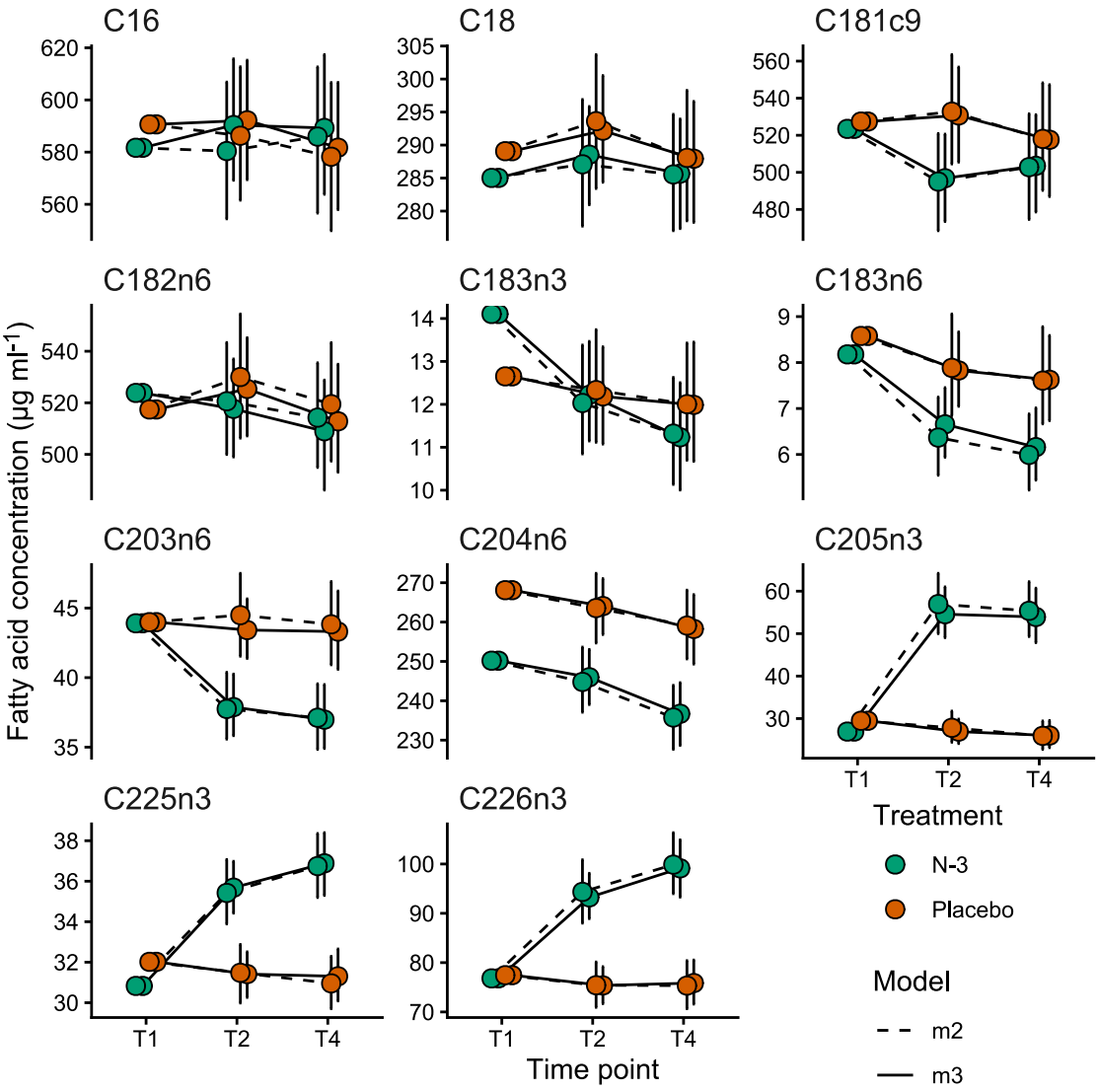


Figure 3.11: Posterior predictions for fatty acid concentrations in complete-case analysis (model m2) and data imputation model (model m3).

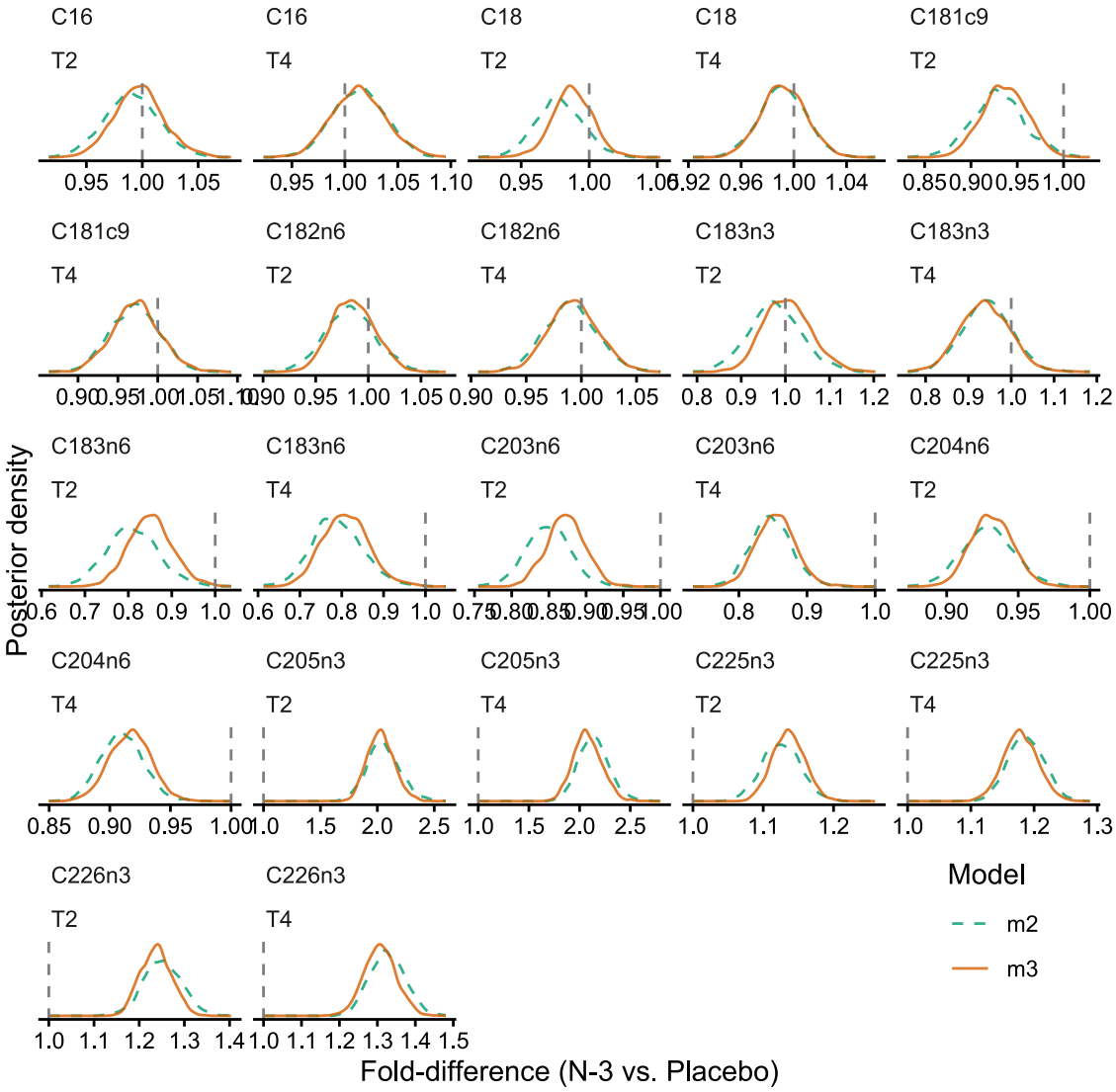


Figure 3.12: Comparing inference from complete-case and imputation models.

3.0.1 Model expansion: Heterogeneous variance

Model `m3` does not enable treatment- and time-wise heterogeneous variation. In model `m4` we therefore add $\sigma_{[\text{FA, TX}]} \sim \text{Exponential}\ (1)$, in model `m5` we add $\sigma_{[\text{FA, TI}]} \sim \text{Exponential}\ (1)$, and model `m6` adds $\sigma_{[\text{FA, TI, TX}]} \sim \text{Exponential}\ (1)$.

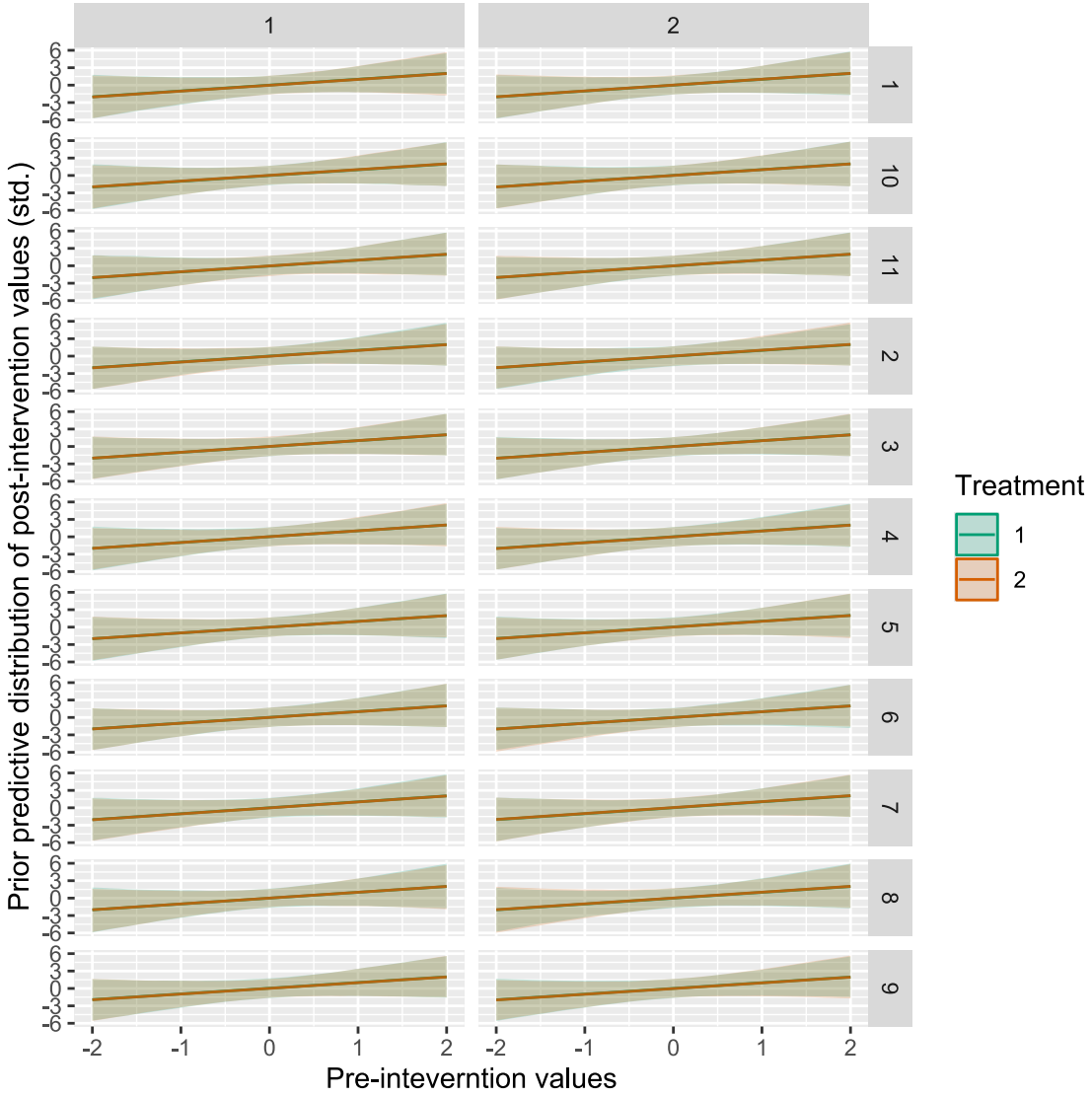


Figure 3.13: Prior predictive check of Models 4-6.

```
Running MCMC with 4 parallel chains...

Chain 1 Iteration:    1 / 3000 [  0%]  (Warmup)
```

```
Chain 2 Iteration:    1 / 3000 [  0%]  (Warmup)
```

```
Chain 3 Iteration:    1 / 3000 [  0%]  (Warmup)
```

```
Chain 4 Iteration:    1 / 3000 [  0%]  (Warmup)
```

```
Chain 1 Iteration: 1000 / 3000 [ 33%]  (Warmup)
Chain 1 Iteration: 1001 / 3000 [ 33%]  (Sampling)
Chain 2 Iteration: 1000 / 3000 [ 33%]  (Warmup)
Chain 3 Iteration: 1000 / 3000 [ 33%]  (Warmup)
Chain 4 Iteration: 1000 / 3000 [ 33%]  (Warmup)
Chain 2 Iteration: 1001 / 3000 [ 33%]  (Sampling)
```

```
Chain 3 Iteration: 1001 / 3000 [ 33%]  (Sampling)
Chain 4 Iteration: 1001 / 3000 [ 33%]  (Sampling)
Chain 1 Iteration: 2000 / 3000 [ 66%]  (Sampling)
Chain 3 Iteration: 2000 / 3000 [ 66%]  (Sampling)
Chain 2 Iteration: 2000 / 3000 [ 66%]  (Sampling)
Chain 4 Iteration: 2000 / 3000 [ 66%]  (Sampling)
Chain 1 Iteration: 3000 / 3000 [100%]  (Sampling)
Chain 3 Iteration: 3000 / 3000 [100%]  (Sampling)
Chain 1 finished in 29.8 seconds.
Chain 3 finished in 29.8 seconds.
Chain 2 Iteration: 3000 / 3000 [100%]  (Sampling)
Chain 2 finished in 29.9 seconds.
Chain 4 Iteration: 3000 / 3000 [100%]  (Sampling)
Chain 4 finished in 29.9 seconds.

All 4 chains finished successfully.
Mean chain execution time: 29.9 seconds.
Total execution time: 30.2 seconds.
```

```
Running MCMC with 4 parallel chains...

Chain 1 Iteration:    1 / 3000 [  0%]  (Warmup)
```

```
Chain 2 Iteration:    1 / 3000 [  0%]  (Warmup)
```

```
Chain 3 Iteration:    1 / 3000 [  0%]  (Warmup)
```

```
Chain 4 Iteration:    1 / 3000 [  0%]  (Warmup)
```

```
Chain 4 Iteration: 1000 / 3000 [ 33%]  (Warmup)
Chain 1 Iteration: 1000 / 3000 [ 33%]  (Warmup)
Chain 3 Iteration: 1000 / 3000 [ 33%]  (Warmup)
Chain 4 Iteration: 1001 / 3000 [ 33%]  (Sampling)
Chain 1 Iteration: 1001 / 3000 [ 33%]  (Sampling)
Chain 2 Iteration: 1000 / 3000 [ 33%]  (Warmup)
Chain 3 Iteration: 1001 / 3000 [ 33%]  (Sampling)
Chain 2 Iteration: 1001 / 3000 [ 33%]  (Sampling)
Chain 4 Iteration: 2000 / 3000 [ 66%]  (Sampling)
Chain 1 Iteration: 2000 / 3000 [ 66%]  (Sampling)
Chain 3 Iteration: 2000 / 3000 [ 66%]  (Sampling)
Chain 2 Iteration: 2000 / 3000 [ 66%]  (Sampling)
Chain 4 Iteration: 3000 / 3000 [100%]  (Sampling)
Chain 4 finished in 28.7 seconds.
Chain 3 Iteration: 3000 / 3000 [100%]  (Sampling)
Chain 3 finished in 28.9 seconds.
Chain 1 Iteration: 3000 / 3000 [100%]  (Sampling)
Chain 2 Iteration: 3000 / 3000 [100%]  (Sampling)
Chain 1 finished in 29.0 seconds.
Chain 2 finished in 29.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 28.9 seconds.
Total execution time: 29.1 seconds.
```

```
Running MCMC with 4 parallel chains...

Chain 1 Iteration:    1 / 3000 [  0%]  (Warmup)
```

```
Chain 2 Iteration:    1 / 3000 [  0%]  (Warmup)
```

```
Chain 3 Iteration:    1 / 3000 [  0%]  (Warmup)
```

```
Chain 4 Iteration:    1 / 3000 [  0%]  (Warmup)
```

```
Chain 1 Iteration: 1000 / 3000 [ 33%]  (Warmup)
Chain 1 Iteration: 1001 / 3000 [ 33%]  (Sampling)
Chain 2 Iteration: 1000 / 3000 [ 33%]  (Warmup)
Chain 2 Iteration: 1001 / 3000 [ 33%]  (Sampling)
Chain 4 Iteration: 1000 / 3000 [ 33%]  (Warmup)
Chain 4 Iteration: 1001 / 3000 [ 33%]  (Sampling)
Chain 3 Iteration: 1000 / 3000 [ 33%]  (Warmup)
Chain 3 Iteration: 1001 / 3000 [ 33%]  (Sampling)
Chain 1 Iteration: 2000 / 3000 [ 66%]  (Sampling)
Chain 4 Iteration: 2000 / 3000 [ 66%]  (Sampling)
Chain 2 Iteration: 2000 / 3000 [ 66%]  (Sampling)
Chain 3 Iteration: 2000 / 3000 [ 66%]  (Sampling)
Chain 1 Iteration: 3000 / 3000 [100%]  (Sampling)
Chain 4 Iteration: 3000 / 3000 [100%]  (Sampling)
Chain 1 finished in 51.4 seconds.
Chain 4 finished in 51.4 seconds.
Chain 2 Iteration: 3000 / 3000 [100%]  (Sampling)
Chain 2 finished in 53.3 seconds.
Chain 3 Iteration: 3000 / 3000 [100%]  (Sampling)
Chain 3 finished in 55.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 52.8 seconds.
Total execution time: 55.2 seconds.
```

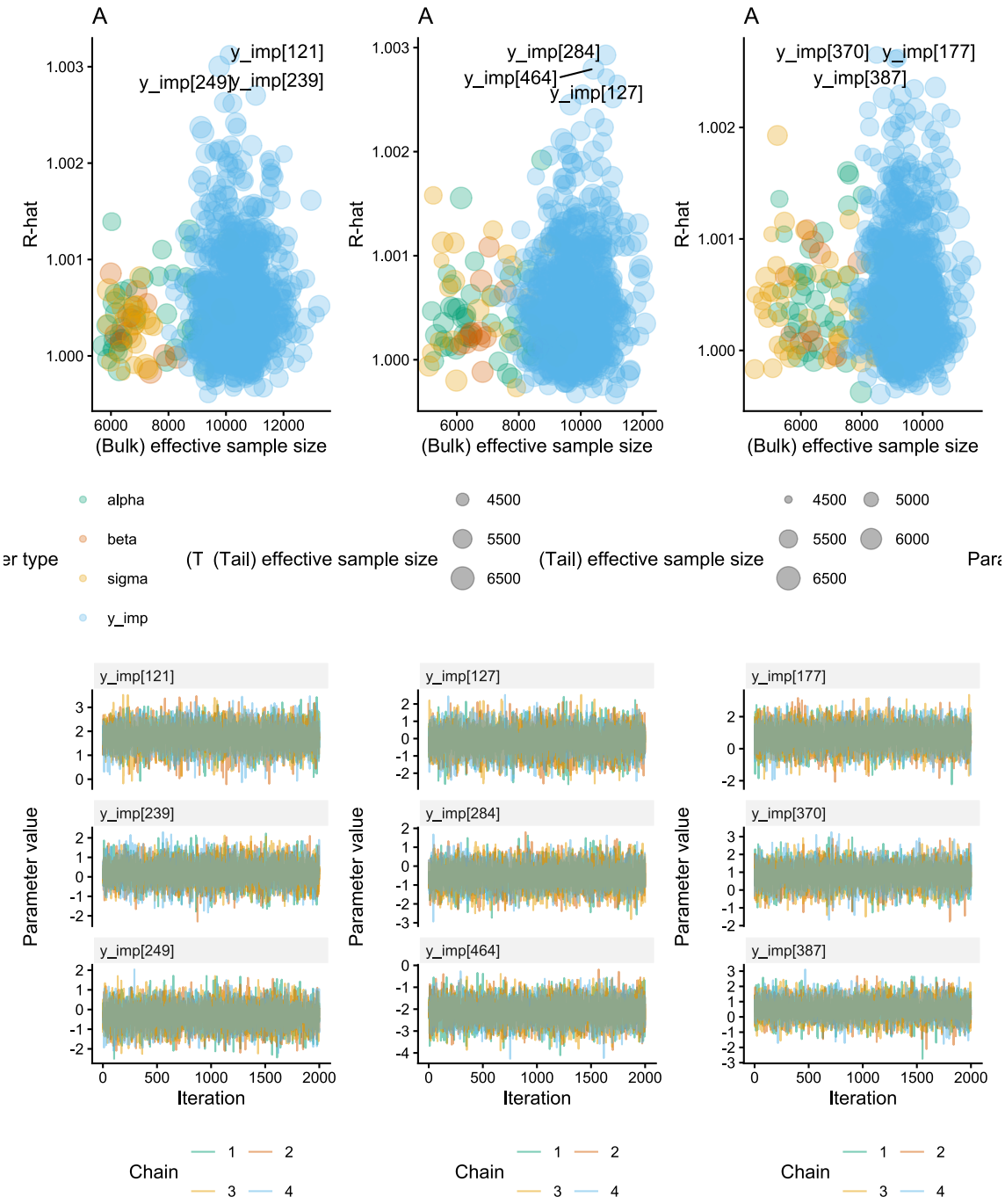


Figure 3.14: Model performance metrics models 4, 5, and 6.

3.1 Inference

4. Effects of N-3 supplementation on indications of metabolic syndrome - Introduction

4.1 Effects of N-3 supplementation on indicies of metabolic syndrome

Here we aim to model the sub-scores for the metabolic syndrome diagnostic criteria, along with related outcomes (Table 4.2). The goal is to build a model that imputes missing data to enable an intention-to-treat (ITT) analysis of all randomized participants. Imputation will allow for assessing the probability of meeting criteria thresholds for having at least 3 of 5 diagnostic criteria for metabolic syndrome in each treatment group while accounting for uncertainty associated with missing observations. Figure 4.1 shows distributions of the variables selected for joint modeling, including variables used to define metabolic syndrome.

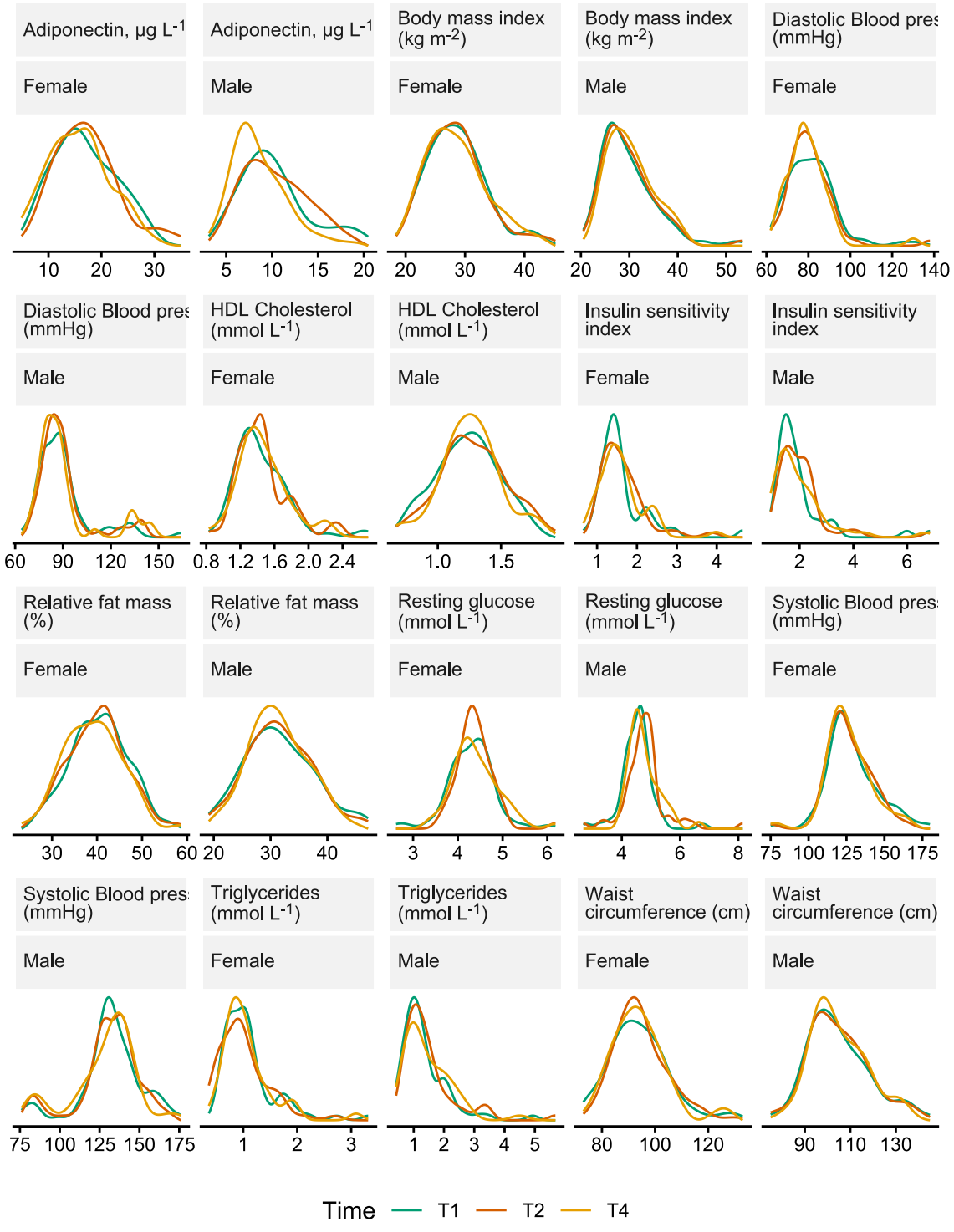


Figure 4.1: Observed data distributions of main outcomes.

Descriptive analysis of complete cases indicates variation in the proportion of participants meeting the criteria thresholds across treatment groups and post-intervention time points. Table 4.1 and Figure 4.2 show the proportions of participants who have three or more diagnostic criteria for metabolic syndrome among complete cases.

Table 4.1: Participants in the complete-case analysis of metabolic syndrome indications.

	Time		
	T1	T2	T4
	Male		
Placebo	21 (19%)	21 (29%)	21 (38%)
N-3	19 (32%)	19 (16%)	19 (32%)
	Female		
Placebo	16 (12%)	16 (12%)	16 (19%)
N-3	21 (19%)	21 (10%)	21 (14%)
	Total		
Placebo	37 (16%)	37 (22%)	37 (30%)
N-3	40 (25%)	40 (12%)	40 (22%)

Data are presented as n (Metabolic syndrome proportion)

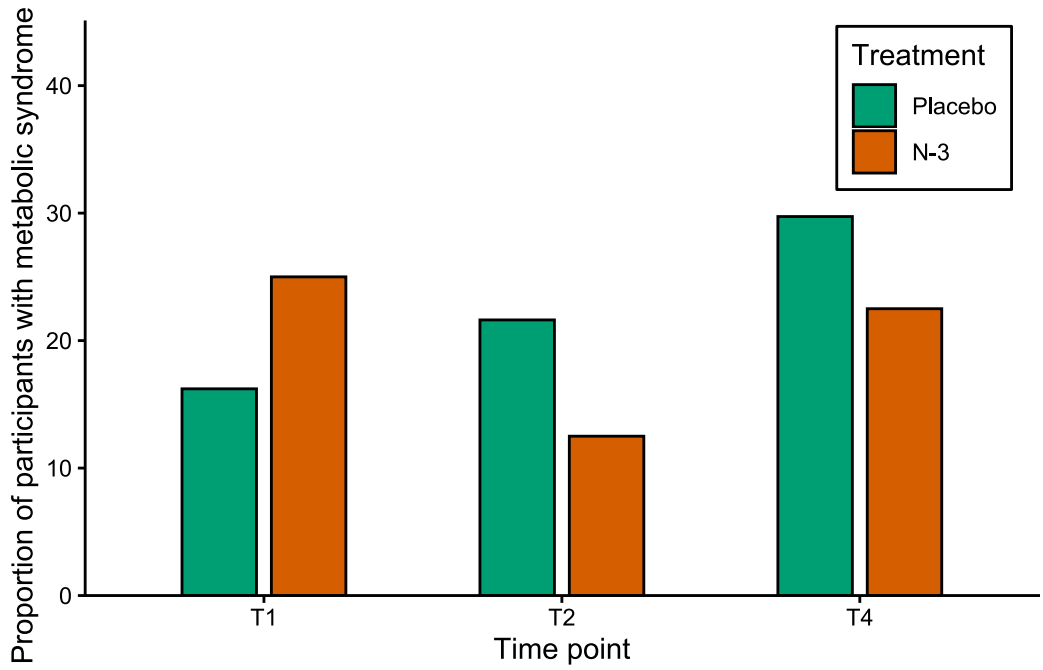


Figure 4.2: Proportion of participants with metabolic syndrome (at least three criteria thresholds).

Table 4.2: Components of the metabolic syndrome diagnostic criteria and related measurements used in modelling.

Subscore	Threshold
HDL Cholesterol	< 40 mg/dL for men, < 50 mg/dL for women
Fasting glucose	100 mg/dL or higher (or taking medication)
Triglycerides	150 mg/dL or higher (or taking medication)
Waist circumference	> 102 cm for men, > 88 cm for women
Blood pressure	130/85 mm Hg or higher (or taking medication)
Relative fat mass	NA
Body mass index	NA
Insulin resistance index	NA
Adiponectin	NA

Variables in the observed data are correlated within time points within variables and between variables across time points. To describe approximate variation and co-variation patterns, "naive" linear models were fitted with sex as a single predictor at time-point T1, sex, baseline values, and treatment as predictors for values at time-points T2 and T4. Residuals were extracted from the models, and estimates were plotted in Figure 4.3.

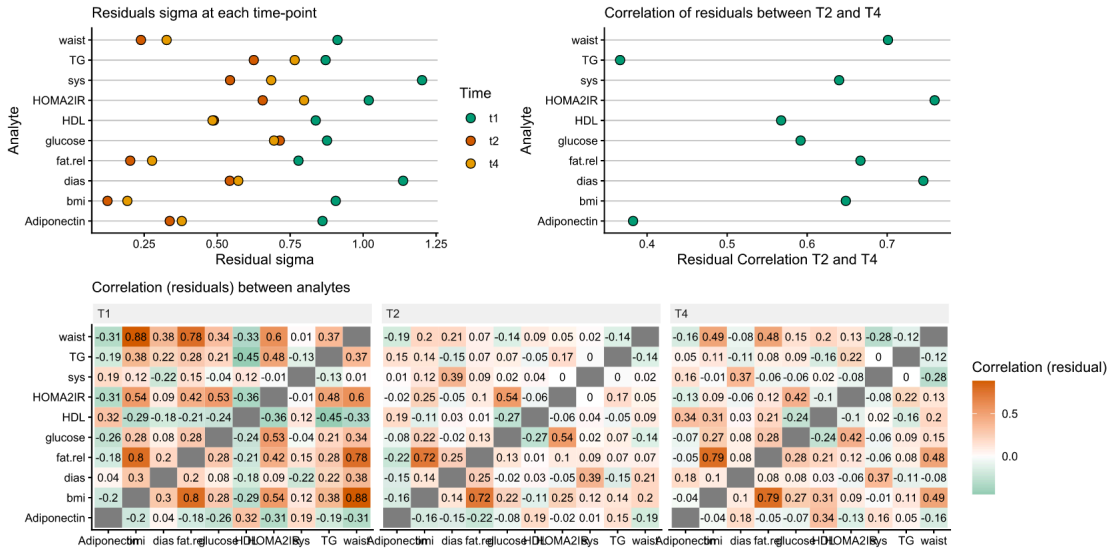


Figure 4.3: Residual variation and correlation pattern within and between variables in the observed data.

Correlation patterns among variables will be used to inform simulations (see Section 4.1.1). Correlations among variables also suggest that uncertainty associated with missing data can be reduced by exploiting the covariance among variables. The missing data patterns of the observed data indicate that some missing data occurs in a single variable at a single time point whilst some participants have missing data in all variables at specific time points (Figure 4.4).

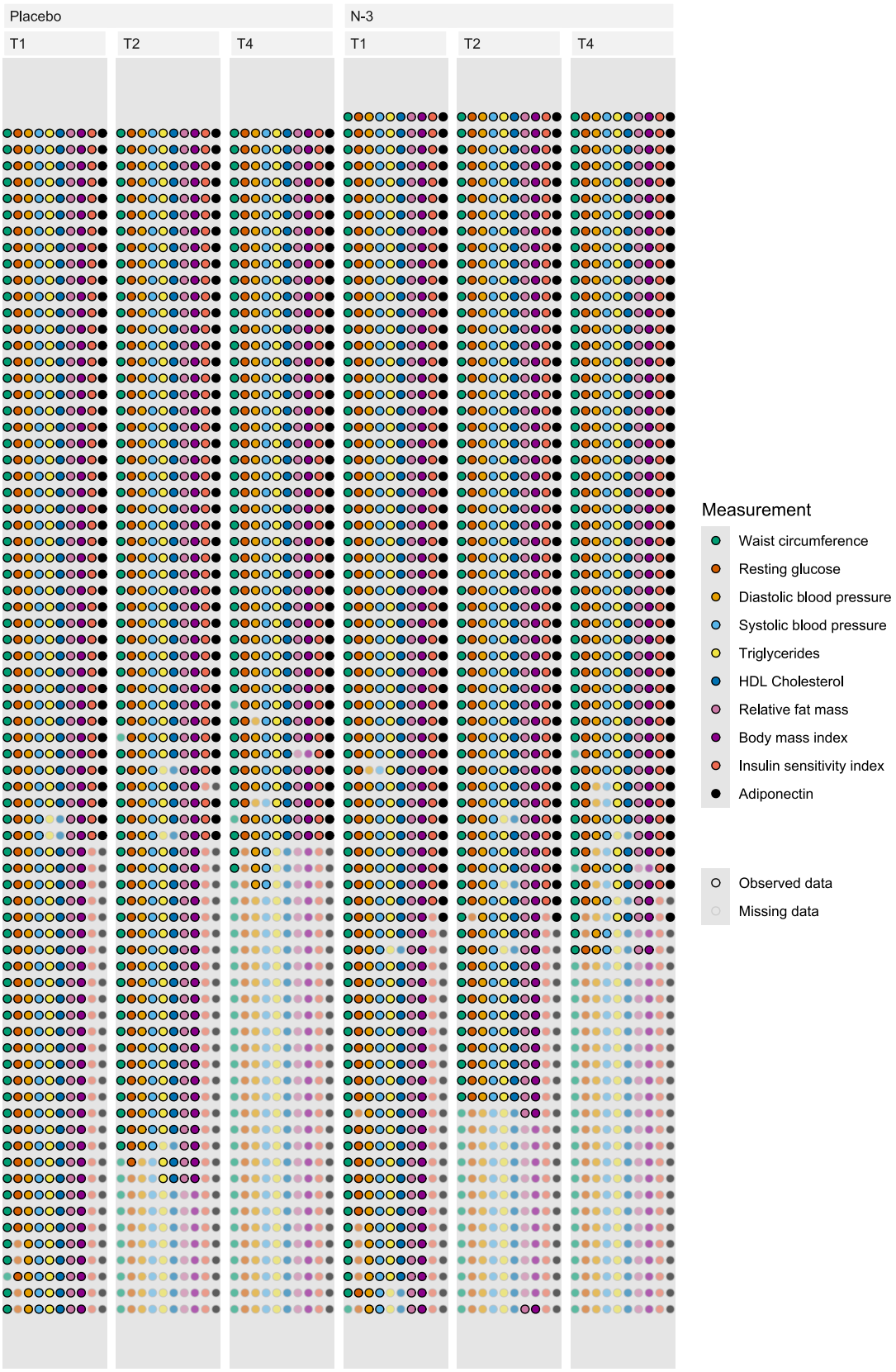


Figure 4.4: Observed and missing data across treatment groups and time-points. Rows are participants, columns are data points.

To assess if the outcome variables predict drop-out we fit a logistic regression model with drop-out (D_i) at time Time T2 or T4, coded as 0/1 as the outcome variable. The association between the standardized measurement type value ($X_{i[A]}$) and drop-out was determined while controlling for sex (S) within each treatment group (T).

$$
\begin{aligned}
D_{i[\text{Time}]} &\sim \text{Bernoulli } (p_i) \\
\text{Logit } (p_i) &= \alpha_{[A]} + \beta_{X[A]} X_{i[A]} + \beta_{S[A]} S_i + \beta_{T[A]} T_i + \beta_{TX[A]} T_i X_i \\
\alpha_{[A]} &\sim \text{Normal } (0, 0.5) \\
\beta_{X[A]} &\sim \text{Normal } (0, 0.5) \\
\beta_{S[A]} &\sim \text{Normal } (0, 0.5) \\
\beta_{T[A]} &\sim \text{Normal } (0, 0.5) \\
\beta_{TX[A]} &\sim \text{Normal } (0, 0.5)
\end{aligned}
\tag{4.1}
$$

In addition to measurements included in the metabolic syndrome diagnostic criteria (Fasting glucose, waist circumference, HDL, triglycerides, blood pressure), we also included relative fat mass, body mass index, age and insulin resistance index (HOMA2IR) and adiponectin. Prior predictive checks (Figure 4.5) confirmed that the model expects eleven measurement types in two treatment groups and that the combined priors weakly regularize away from all/non drop-out.

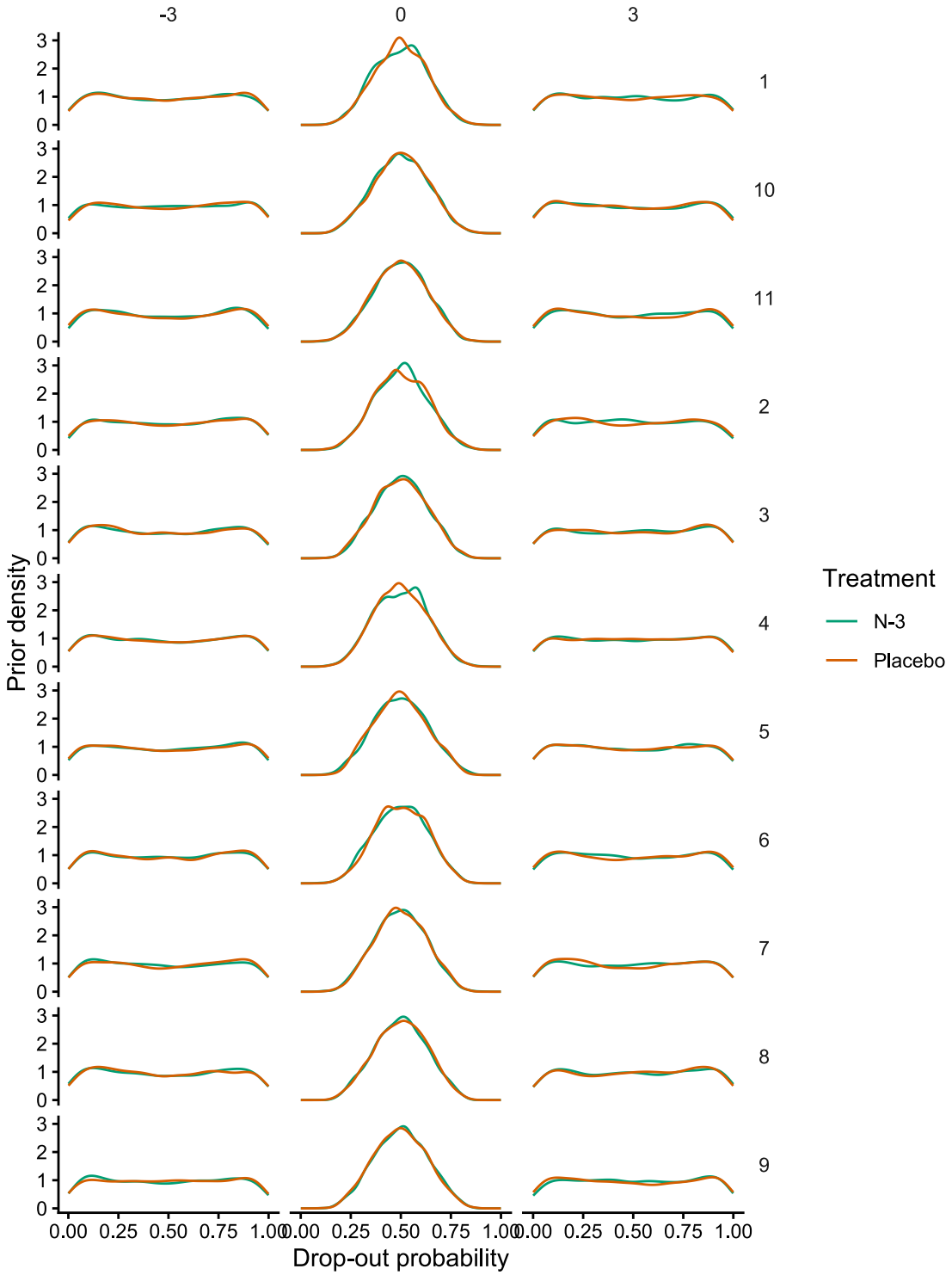


Figure 4.5: Prior predictive simulation for model 1. Columns are values of the predictor (SD-units) rows represents each predictor.

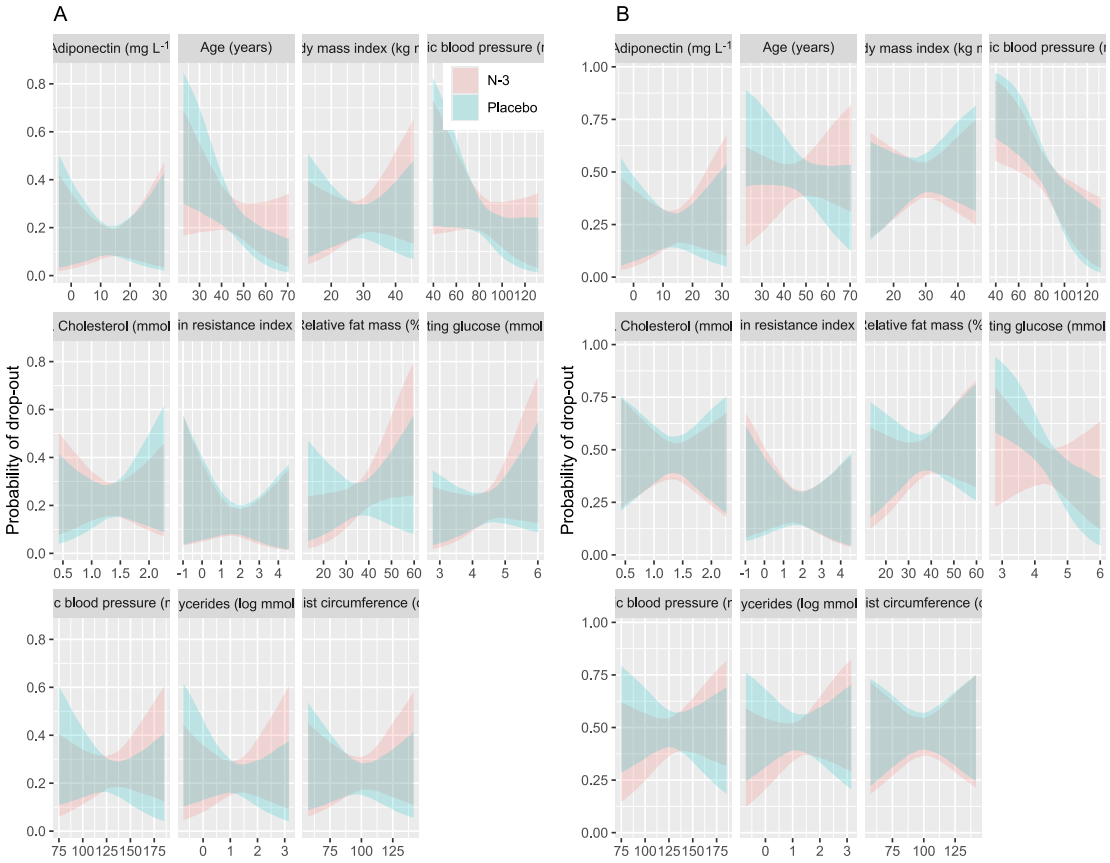


Figure 4.6: Probability of drop-out across values of baseline co-variates on per treatment group to time-point 2 (A) and 4 (B). Estimates are adjusted for sex, intervals are 90% percentile interval.

Table 4.3: Change in log-odds (Slope) for drop-out for 1 SD-unit change in baseline characteristic (Outcome) and differences in slope between treatment groups (&Delta Treatment) at time T2 and time T4. Values are posterior means with 95% percentile intervals.

Outcome	T2		T4	
	Slope	Δ Treatment	Slope	Δ Treatment
Adiponectin (mg L-1)	0.01 [−0.54, 0.54]	0.14 [−0.62, 0.89]	0.07 [−0.39, 0.54]	0.18 [−0.53, 0.89]
HDL Cholesterol (mmol L-1)	0.05 [−0.35, 0.45]	−0.16 [−0.78, 0.45]	−0.03 [−0.37, 0.32]	−0.04 [−0.63, 0.55]
Insulin resistance index (log)	−0.13 [−0.72, 0.42]	−0.00 [−0.80, 0.78]	−0.10 [−0.59, 0.35]	−0.05 [−0.76, 0.67]
Triglycerides (log mmol L-1)	−0.02 [−0.42, 0.37]	0.27 [−0.36, 0.90]	0.07 [−0.26, 0.42]	0.22 [−0.36, 0.81]
Age (years)	−0.41 [−0.80, −0.05]	0.31 [−0.29, 0.92]	−0.07 [−0.39, 0.25]	0.45 [−0.09, 1.03]
Body mass index (kg m-2)	0.08 [−0.29, 0.44]	0.20 [−0.41, 0.82]	0.10 [−0.22, 0.41]	−0.08 [−0.63, 0.47]
Diastolic blood pressure (mmHg)	−0.37 [−0.82, 0.05]	0.17 [−0.50, 0.84]	−0.62 [−1.06, −0.23]	0.18 [−0.48, 0.84]
Relative fat mass (%)	0.25 [−0.14, 0.66]	0.36 [−0.25, 0.99]	0.13 [−0.21, 0.51]	0.13 [−0.45, 0.72]
Resting glucose (mmol L-1)	0.24 [−0.15, 0.66]	0.19 [−0.50, 0.84]	−0.34 [−0.71, 0.01]	0.43 [−0.21, 1.07]
Systolic blood pressure (mmHg)	−0.01 [−0.38, 0.36]	0.28 [−0.29, 0.87]	0.04 [−0.30, 0.36]	0.25 [−0.31, 0.82]
Waist circumference (cm)	0.01 [−0.37, 0.38]	0.19 [−0.39, 0.76]	0.03 [−0.33, 0.35]	0.01 [−0.52, 0.58]

In conclusion, age was negatively associated with probability of drop-out at T2 without robust difference between treatment groups (Figure 4.6 A; Table 4.3). Resting glucose at baseline tended to predict drop-out to T4 with no strong difference between treatment groups (Figure 4.6 B; Table 4.3). Together this suggest that baseline covariates did not robustly interact with treatment in predicting drop-out. However, baseline covariates (age and fasting glucose) were associated with higher drop-out rates.

4.1.1 Synthetic data

Synthetic data is generated with known parameter values to validate models and to explore the effects of model misspecification and missing data. The first data simulation function uses a multivariate normal distribution to simulate conditional means for K measurement types and N participants.

$$\begin{aligned} y_{i,j} &\sim \text{MVNormal}\ (\mu_{k[i,j]}, \Sigma) \\ \mu_{k[i,j]} &= \alpha + \beta_k X \end{aligned} \tag{4.2}$$

Where $y_{i,j}$ is a vector of outcome values for participant i in time-point j, drawn from a multivariate normal distribution with conditional means ($\mu_{k[i,j]}$ and a variance-covariance matrix (Σ). The conditional means are simulated using a linear predictor with parameter vector β and design matrix X.

The covariance matrix is constructed from two separate correlation matrices and a cross-time-point vector of correlations within each measurement type. Residual standard deviations for each type (and

time) are used to combine standard deviations and correlations into the variance-covariance matrix[1].

$$
\begin{aligned}
\rho_{t2} &= \begin{bmatrix} 1 & \rho_{12} & \cdots \\ \rho_{21} & 1 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \\
\rho_{t4} &= \begin{bmatrix} 1 & \rho_{12} & \cdots \\ \rho_{21} & 1 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \\
\rho_{\text{cross}} &= \text{diag } (p_{k,\text{ time}}) \\
R &= \begin{bmatrix} \rho_{t2} & \rho_{\text{cross}} \\ \rho_{\text{cross}}^T & \rho_{t4} \end{bmatrix} \\
S &= \text{diag } (\sigma_{k,\text{ time}}) \\
\Sigma &= S \times R \times S
\end{aligned}
\tag{4.3}
$$

We will be able to control the within-time *residual* correlations between measurement types and between-time *residual* correlations within measurement types. The full correlation matrix (R) will set the residual correlation between measurement types at different time points to 0. These will, however, be correlated through within-time-point and pre-intervention correlations.

The pre-intervention values are also conditionally constructed using a separate multivariate normal distribution.

$$
\begin{aligned}
y_{\text{pre }[i]} &\sim \text{MVNormal } (\mu_{k[i]}, \Sigma) \\
\mu_{k[i]} &= \alpha + \beta_k X \\
R &= \begin{bmatrix} 1 & \rho_{12} & \cdots \\ \rho_{21} & 1 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \\
S &= \text{diag } (\sigma_k) \\
\Sigma &= S \times R \times S
\end{aligned}
\tag{4.4}
$$

Using this definition, we can control the correlations among measurement types (at baseline) and their variation. We can also make the means conditional on a set of predictors (design matrix X) with corresponding parameter values (β_k).

The approach detailed above is convenient as we are able to define each pairwise correlation. However, this comes with two caveats. First, for 8 variables we need to set $((8 \times 8 - 8)/2) \times 3 = 84$ correlation coefficients to capture between variable correlations at each time-point. Second, this approach does not guarantee a positive definite covariance matrix. An alternative is to use a factor loading approach[2]. For each variable we can specify a loading onto a factor and a residual variance for each variable (type). This approach is determined by λ, a $K \times F$ matrix where k_i are variables and f_j are latent factors, and ψ, a vector of length K defining residual variation.

$$
\begin{aligned}
\Sigma_{\text{raw}} &= \lambda \times \lambda^T + \text{diag } (\psi) \\
D &= \sqrt{\text{diag } (\Sigma_{\text{raw}})} \\
R &= D^{-1} \times \Sigma_{\text{raw}} \times D^{-1} \\
S &= \text{diag } (\sigma_k) \\
\Sigma &= S \times R \times S
\end{aligned}
\tag{4.5}
$$

This approach is used in `sim_type2` where Σ is built from the factor approach. Lambda is a matrix with K rows, representing variable types and F columns representing factors.

[1] In the simulation function (`./R/sim-fun/sim-type.R`) separate variance-covariance matrices are constructed and then combined. But separate correlation matrices are the function's input.

[2] See here for an introduction on the maths.

```
build_sigma_factor <- function(Lambda, psi, S) {
  K <- nrow(Lambda)
  R_raw <- Lambda %*% t(Lambda) + diag(psi)
  d <- sqrt(diag(R_raw))
  R <- R_raw / outer(d, d)
  diag(S) %*% R %*% diag(S)
}
```

As an example, $K = 5$ variable types are defined using $F = 3$ factors below. Approximate "variance explained" can be used to define the relationship between variables and factors where for k_1 we aim for 50%, 20% and 10% explained by f_1 to f_3 and the remaining 20% is error. The squared loadings plus error sum to 1: $0.5 + 0.2 + 0.1 + 0.2 = 1$. In code:

```
# Loadings as proportions
l1 <- c(0.5,  0.2,  0.1)
# Transforms to loadings
l1 <- sqrt(l1)
error <- 0.2

sum(l1^2) + error
```

```
[1] 1
```

The above example first defines loadings as proportions (summing to 1 with errors) and then adds a step to transform the proportions to loadings. For a full $K \times F$ λ matrix we populate the matrix and use ψ for errors.

```
# Defining a matrix of proportions explained by factors
Lambda <- matrix(
  c(0.5, 0.2, 0.1,
    0.8, 0.1, 0.05,
    0.1, 0.15, 0.1,
    0.05, 0.05, 0.2,
    0.05, 0.1, 0.01
    ),
                ncol = 3,
                byrow = TRUE)
# Defining remaining error
psi <- c(0.2, 0.05, 0.65, 0.7, 0.84)

# Transform from proportions to loadings
Lambda <- sqrt(Lambda)

# Check total proportions
rowSums(Lambda^2) + psi
```

```
[1] 1 1 1 1 1
```

Each row of the λ matrix plus errors now sum to 1. Next, we can use the λ matrix to construct Σ with unit standard deviations. The full covariance matrix (Σ) is used in calling `MASS::mvnorm` to produce the multivariate normal variable. As we are using standard deviations equal to 1, the covariance matrix equals the correlation matrix.

```
S <- rep(1, 5)
Sigma <- build_sigma_factor(Lambda, psi, S)

out <- MASS::mvrnorm(1000, rep(0, 5), Sigma = Sigma)

colnames(out) <- paste0("k", 1:5)
```

Table 4.4: Correlations used to simulate data built using a factor approach.

	k1	k2	k3	k4	k5
k1	1.000	0.845	0.497	0.400	0.331
k2	0.845	1.000	0.476	0.371	0.322
k3	0.497	0.476	1.000	0.299	0.225
k4	0.400	0.371	0.299	1.000	0.165
k5	0.331	0.322	0.225	0.165	1.000

Table 4.5: Correlations from simulated data (n=1000) built using a factor approach.

term	k1	k2	k3	k4	k5
k1	NA	0.854	0.477	0.427	0.313
k2	0.854	NA	0.464	0.438	0.313
k3	0.477	0.464	NA	0.290	0.196
k4	0.427	0.438	0.290	NA	0.144
k5	0.313	0.313	0.196	0.144	NA

In the loadings we defined k_1 and k_2 to be highly correlated through being influenced to a large degree by factors f_1 and f_2. In contrast, k_5 was given a lot of variation not explained by the factors which resulted in lower correlations with the other variables.

4.1.1.1 Heteroscedasticity and non-normal noise

The synthetic data-generating models described above produces homoscedastic data, i.e., equal residual variation between treatment groups. To create synthetic data under heteroscedastic conditions (unequal variance between treatment groups) an additional data generating model is created to enable setting treatment-specific variation (`sim_type3` function in `R/sim-fun.R`). The `sim_type3` function also includes two arguments to set additional noise in post-treatment measurements `prop_noise` controls the fraction of measurements that gets extra noise added and `sd_noise` controls the zero-centered normal distribution from which additional noise is drawn.

4.1.1.2 Missing data

Two mechanisms for including missing values are used in the simulation functions. First, a proportion of pre- and post-intervention values are set to missing, affecting single observations (e.g., a single measurement within a single participant). Second, the probability of missing observations from a whole participant is simulated using a Bernoulli model. This mechanism can connect baseline predictors to missing values. A Bernoulli simulation is performed, with p_i modeled linearly by α, which controls the average probability of complete missing observations, and the additional parameters β linked to predictors in a predictor data frame.

$$\begin{aligned} y_{\text{miss }[i]} &\sim \text{Bernoulli }(p_i) \\ \text{logit }(p_i) &= \alpha + \beta X \end{aligned} \tag{4.6}$$

Simulating a full (K = 8) data set of synthetic data, with missing values, results in observed patterns similar to the observed data (Figure 4.4, Figure 4.7 and Figure 4.8).

```
# A tibble: 3,552 × 8
   participant  type group   sex   age time       y    ystd
         <int> <int> <int> <int> <dbl> <chr>  <dbl>   <dbl>
```

```
 1           1     1     1     1  1.23 T1    -1.46  -1.23
 2           1     1     1     1  1.23 T2    -0.537 -0.568
 3           1     1     1     1  1.23 T3    -0.364 -0.444
 4           1     2     1     1  1.23 T1     0.762  0.524
 5           1     2     1     1  1.23 T2     0.114 -0.0497
 6           1     2     1     1  1.23 T3     1.51   1.19
 7           1     3     1     1  1.23 T1    -1.39  -1.17
 8           1     3     1     1  1.23 T2    -0.647 -0.547
 9           1     3     1     1  1.23 T3    -1.03  -0.869
10           1     4     1     1  1.23 T1     0.400  0.178
# ℹ 3,542 more rows
```

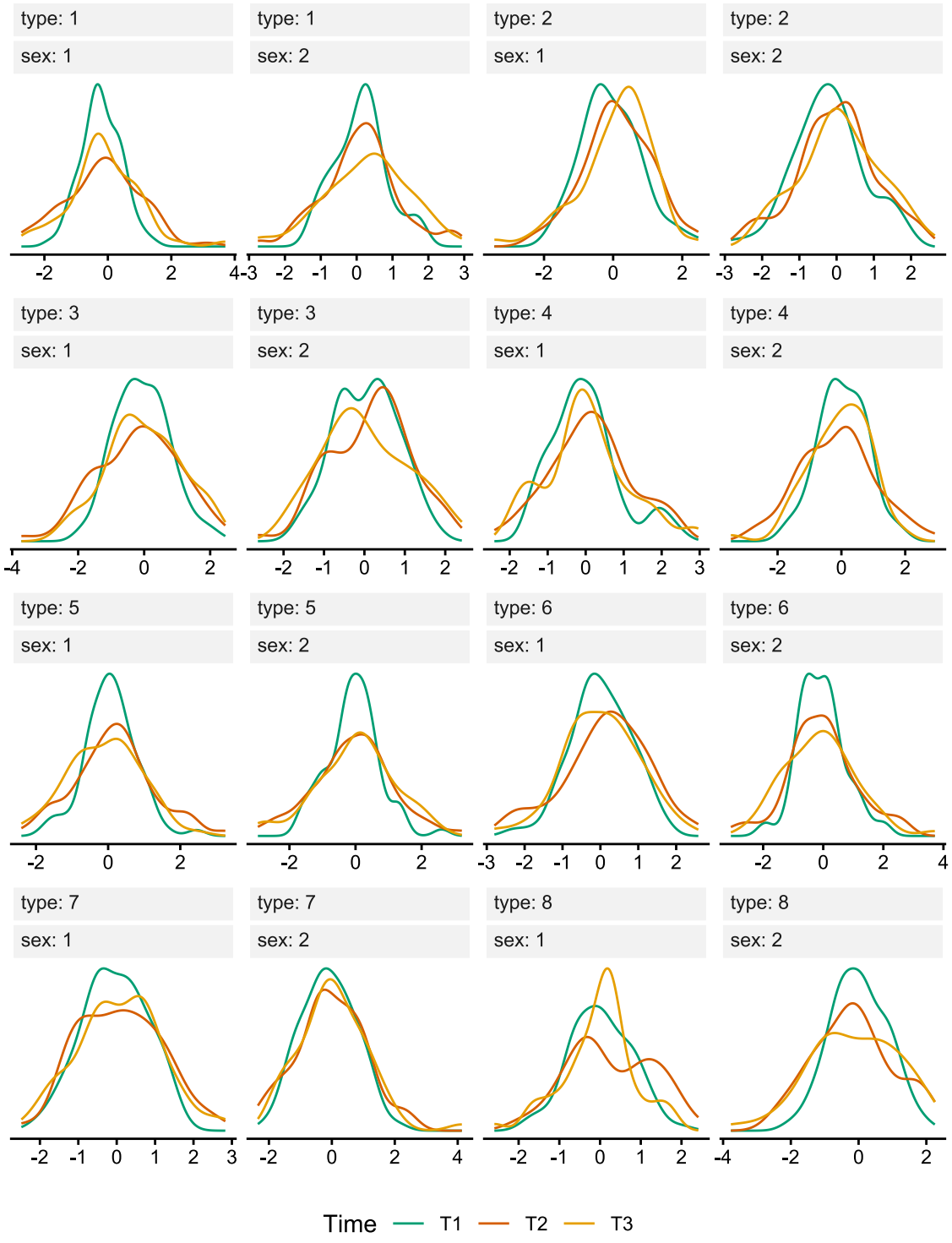


Figure 4.7: Distribution of (standardized) synthetic data across sex and measurement types.

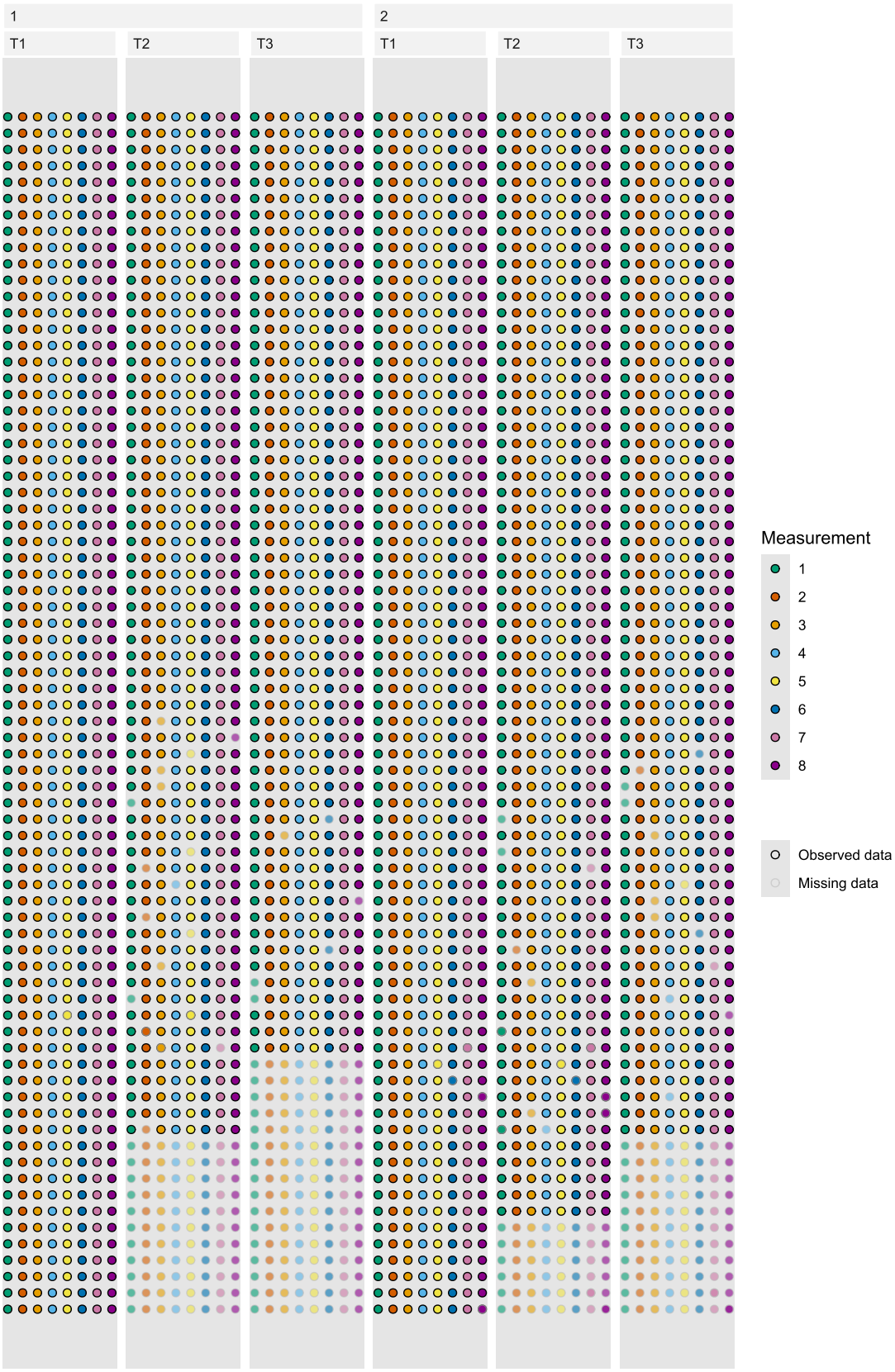


Figure 4.8: Missing data patterns from the data simulation. Missing values are found in specific measurements and in all measurements for a participant at specific time points.

Bibliography

1. Carpenter B, Gelman A, Hoffman MD, Lee D, Goodrich B, Betancourt M, et al. Stan: A Probabilistic Programming Language. Journal of Statistical Software [Internet]. 2017 ;76(1):1–32. Available from: https://www.jstatsoft.org/index.php/jss/article/view/v076i01
2. Gabry J, Cesnovar R, Johnson A, Bronder S. cmdstanr: R Interface to 'CmdStan' [Internet]. 2025. Available from: https://mc-stan.org/cmdstanr/
3. R Core Team. R: A Language and Environment for Statistical Computing [Internet]. Vienna, Austria; 2025. Available from: https://www.r-project.org/
4. Lemoine NP. Moving beyond Noninformative Priors: Why and How to Choose Weakly Informative Priors in Bayesian Analyses. Oikos. 2019 July ;128(7):912–28.
5. McElreath R. Statistical Rethinking: A Bayesian Course with Examples in R and Stan. Second edition. Boca Raton: CRC Press; 2020. (Chapman & Hall/CRC Texts in Statistical Science Series).
6. Gelman A, Carlin JB, Stern HS, Dunson DB, Vehtari A, Rubin DB. Bayesian Data Analysis. Third edition. Boca Raton: CRC Press; 2014.